



GE VERNOVA

ADMS

16.25.1

Geographic Background Compiler User Guide

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Change History

Date	Change Description
May 31, 2023	ADMS 3.16: • Editorial and formatting updates.
Oct 20, 2023	ADMS 16.1.0: • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
Mar 25, 2024	ADMS 16.7.0: • Updated document format and styling.
Mar 3, 2025	ADMS 16.17.0: • Added a Note to Structure Search Display.

Contents

1. Introduction	5
1.1. Geographic Background Compiler	5
1.2. Geographic Background Compiler Configuration File	7
1.3. Import Format	8
2. Displays Overview	10
2.1. Compiler Tab	10
2.2. Log Tab	11
2.3. Tile Preview Tab	11
2.4. Area Preview Tab	12
2.5. Landbase Search Tab	13
2.6. Configuration Window	14
2.6.1. Options Tab	15
2.6.2. Column Names Tab	16
2.6.3. Paths Tab	16
2.6.4. Word Exclusion Tab	17
2.6.5. Feature Mapping Tab	18
3. General Usage	20
3.1. Starting the Geographic Background Compiler	20
3.2. Loading a Geographic Background Compiler Configuration File	20
3.3. Saving a Geographic Background Compiler Configuration File	21
3.4. Assigning Type Codes to Landbase Codes	21
3.5. Compiling Landbase Folders	21
3.6. Viewing Geographic Layers on the Tile Preview Tab	22
3.7. Viewing Geographic Layers on the Area Preview Tab	22
3.8. Performing Command-Line Operations	23
3.8.1. Command-Line Arguments	23
3.8.2. Starting the Geographic Background Compiler from the Command Prompt	24
3.8.3. Compiling a Single Landbase Folder from the Command Prompt	24
3.8.4. Generating a Log File from a Command-Line Start	25
3.8.5. Creating and Running a Background Compiler Build Script	25
3.9. Changing the Default Bitmap Folder Location	27
4. Applying the New Background Data	29
4.1. Generating Background Layers	29
4.2. Defining Background Parameters in the Net Dynamics Configuration Editor	33
4.3. Defining a New Background Layer in the Viewer Configuration Editor	36
4.4. Adding a New Background Layer Label to the Geographic_OpenSpace (Common) Toolbar	38
4.5. Adding a New Landbase Code to the Viewer Configuration Editor	39
5. Modeling Structures	41
5.1. Data Format	42
5.1.1. Tiles	43

5.1.2. DNOM Integration	43
5.2. Indexing	43
5.3. Distribution to Clients	44
5.4. Viewer Configuration	45
5.4.1. Structure Style Parameters	45
5.4.2. Attribute Displays	47
5.4.3. Device Toolbar	47
5.4.4. Geographic Open Space Toolbar Menu	48
5.5. Structure Search Display	48
5.6. Characteristics of Structure Types	49
5.7. Modeling Equipment and Structures	51
5.7.1. Data Format	51
5.7.2. Indexing	53
6. Building the Postal_Inventory.xml File	55
7. The AddressLookup.xml File	56
8. Displaying the Landbase Search Tab	58
9. Landbase Layer Types	59

1. Introduction

The Geographic Background Compiler processes type codes, landbase information that is extracted from the Geographic Information System (GIS) and other sources. The Geographic Background Compiler is a Microsoft Windows application that runs on the model update machine together with the converter.

The Background Compiler produces background files that are incorporated into the model management process. As part of the model management process, the landbase-related files are moved to the model staging machine as soon as they have been approved and deemed ready to put online. From the model staging machine, the files (such as Station Model files) are then moved to each individual ADMS client machine by the File Updater software.

Geographic Background Compiler: The Background Compiler processes GIS landbase information. It then converts that information into a format that helps optimize display call-up as well as intelligent search capabilities.

Geographic Background Compiler Configuration File: The Background Compiler uses a configuration file to manage all the configurable properties.

Geographic Background Compiler Command Line: The Background Compiler is capable of running interactively from the command line based on command-line arguments.

1.1. Geographic Background Compiler

The Geographic Background Compiler processes the type codes that exist in the different shape files from the GIS. It then compiles those shape files into background files that are useable by the ADMS client, as shown in the figure below:

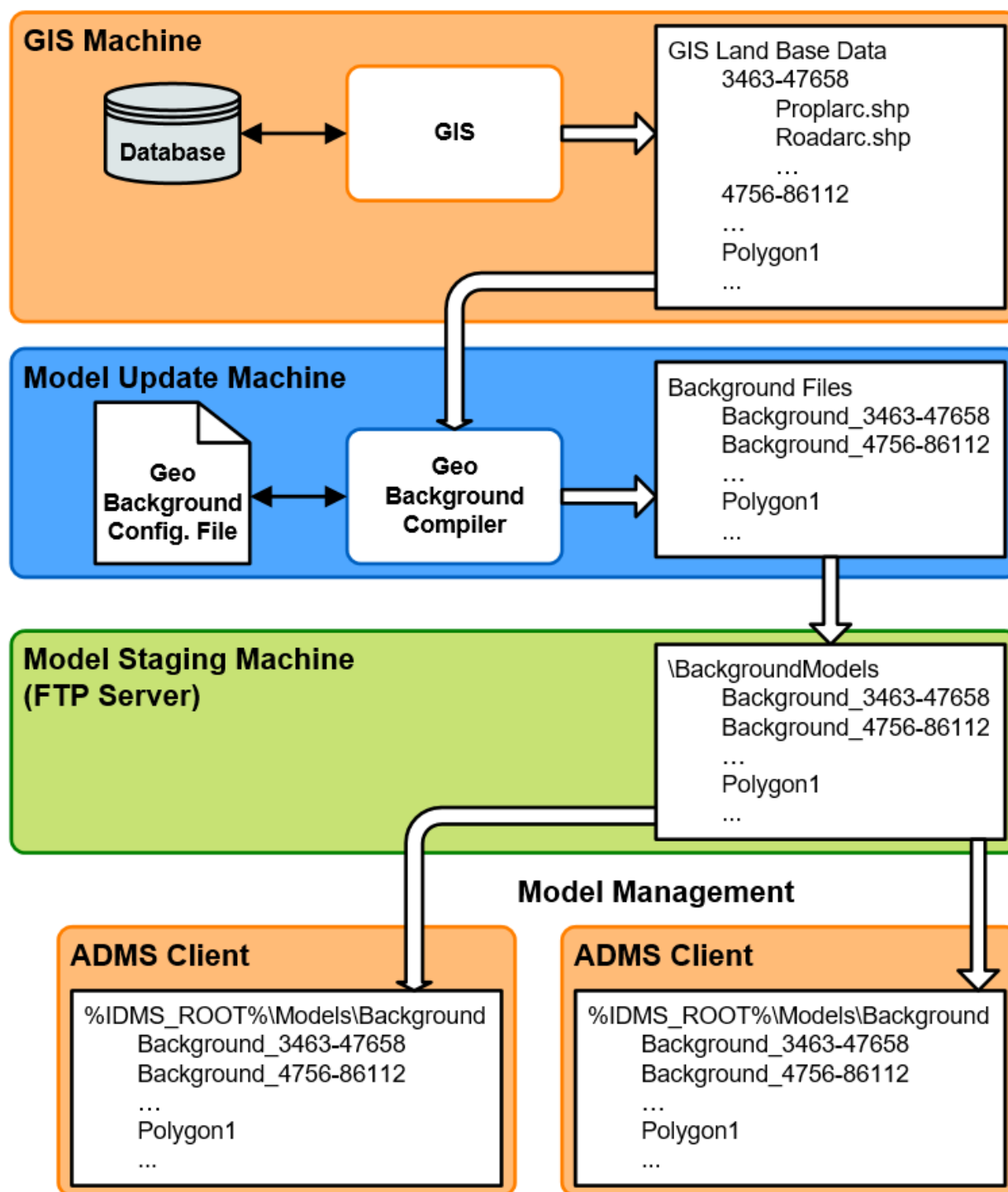


Figure 1. Landbase Data Flow

1.2. Geographic Background Compiler Configuration File

All the configurations specified in the Geographic Background Compiler can be saved in a *.config configuration file. The configuration file can be saved from the File menu.

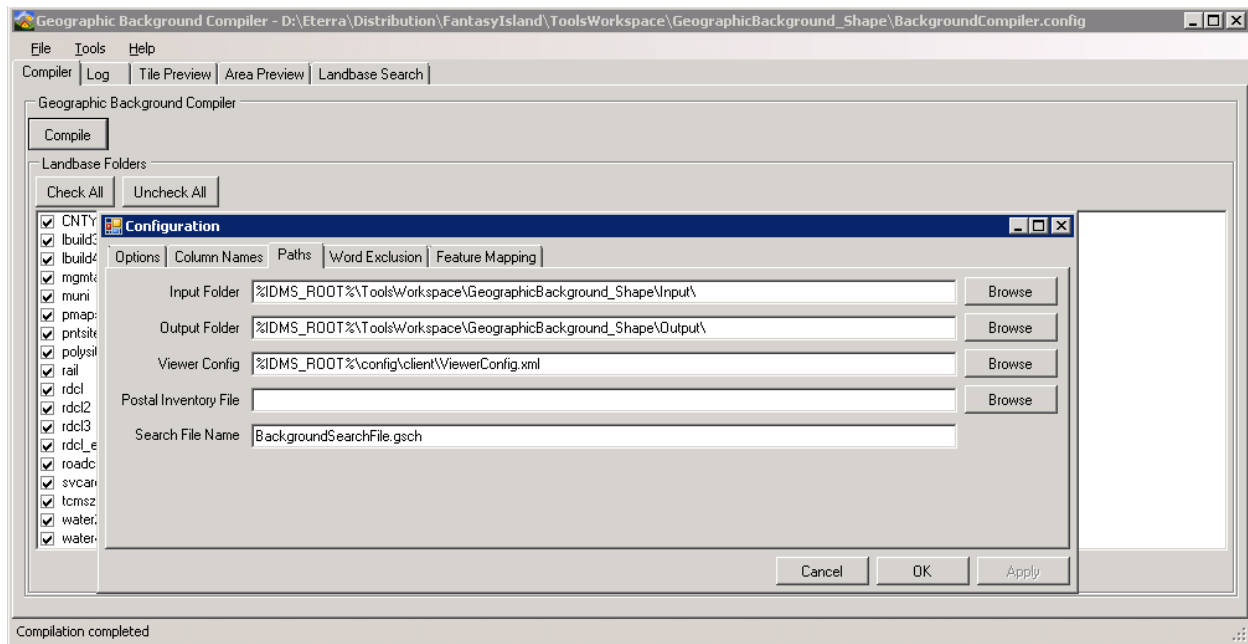


Figure 2. Configuration Window - Paths Tab

You can specify the name and location of the configuration file using the argument "-ConfigFile" when running the Geographic Background Compiler from the command line. It can also be used in shortcuts to set the context when running the Geographic Background Compiler interactively.

Optionally, you can open a Geographic Background Compiler configuration file from the File > Open menu.

The Geographic Background Compiler configuration file contains the mapping of a GIS layer code to a layer name, as shown in the below image.

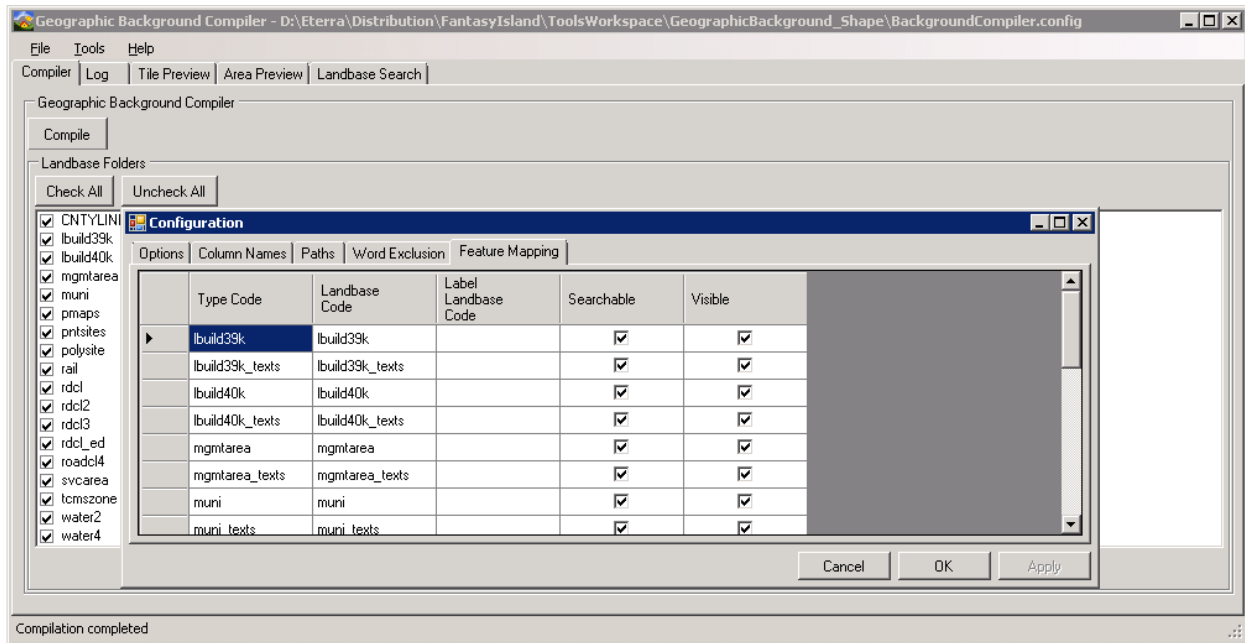


Figure 3. Configuration Window - Feature Mapping Tab

The actual mapping of this layer name (for example, Lake) to a layer category (for example, water) is done in the Viewer Configuration Editor.

1.3. Import Format

The Geographic Background Compiler uses the folder structure provided by GIS, such that top-level directories contain folders that correspond to grid names and polygon names.

The Compiler tab allows you to interactively compile one or many landbase folders. The below image shows a screenshot of this tab.

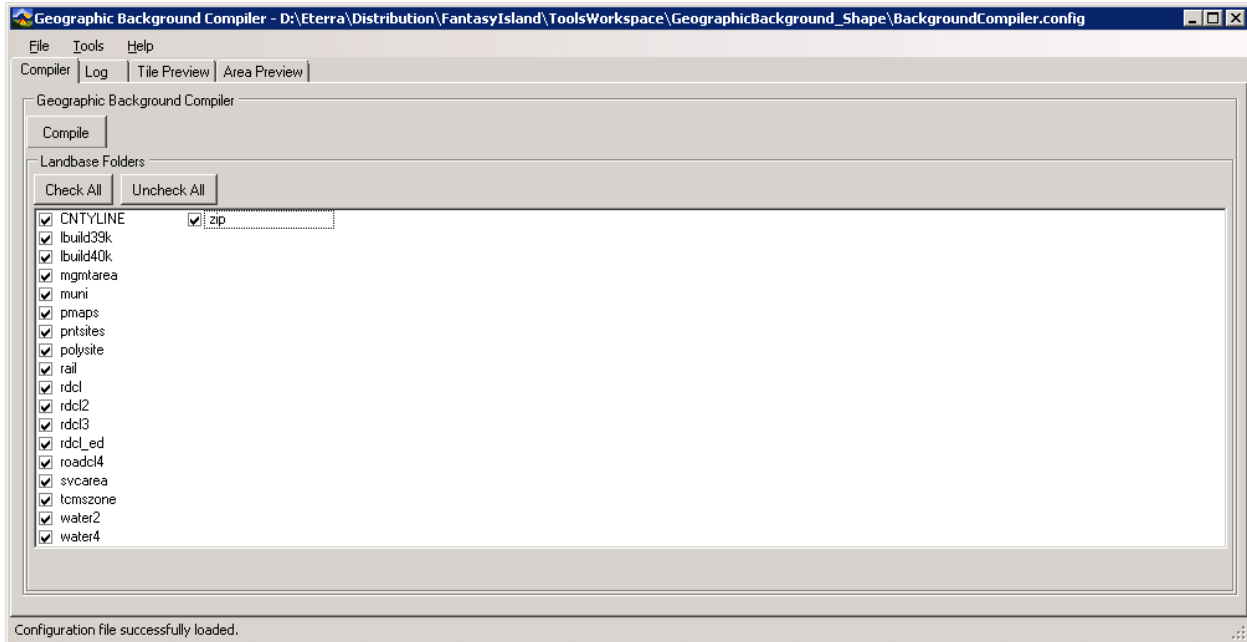


Figure 4. Geographic Background Compiler – Compiler Tab

The Compiler can be configured to use a certain input folder. If properly configured, the Geographic Background Compiler processes the input folder and shows a list of all folders found in it.

You can select one or more folders to compile by selecting its check box, or you can click the Check All button to select all folders.

Note

The Geographic Background Compiler does not support multi-polygon objects.

2. Displays Overview

This chapter describes the tabbed displays of the Geographic Background Compiler.

2.1. Compiler Tab

The Compiler tab (the below image) allows you to interactively compile one or many landbase folders.

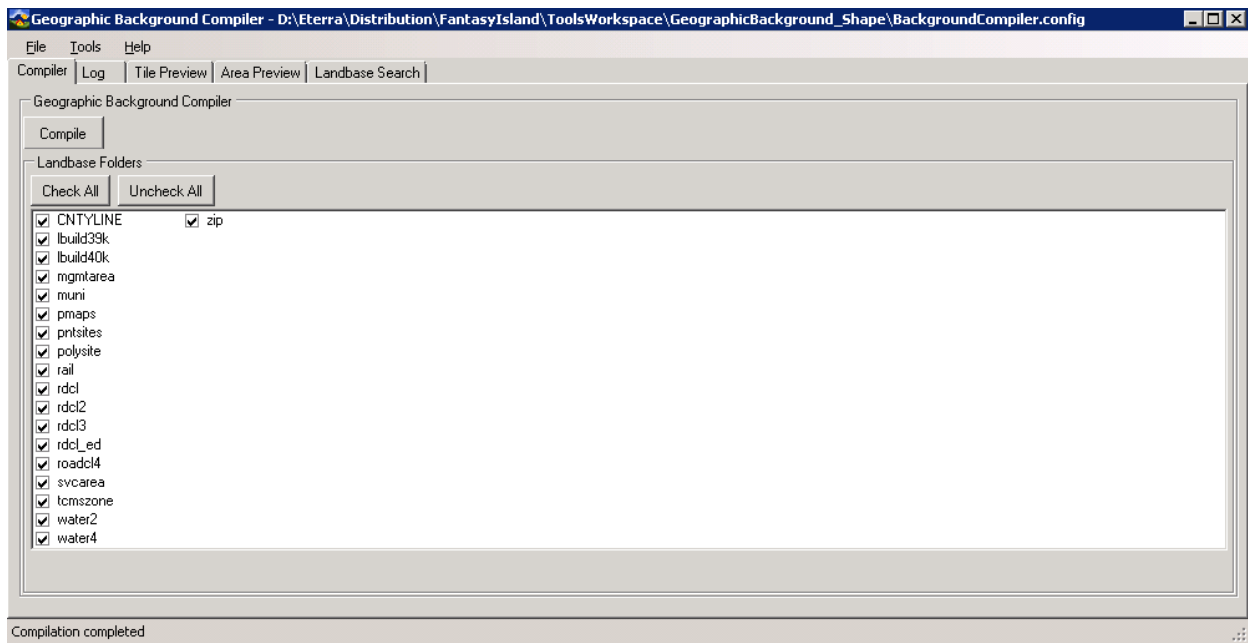


Figure 5. Compiler Tab

The Compiler tab's parameters and fields include:

- **Compile button:** Click to compile the selected Landbase Folders. It also produces a background search file if the Compile Search check box is selected on the Options tab. The Compile button performs the same function as the Compile Background command on the File menu.
- **Landbase Folders:** Displays subfolders of the input folder containing landbase data.
- **Check All button:** Click to select all the folders to compile.
- **Uncheck All button:** Click to clear all the folders to compile.
- **Landbase Folders item check boxes:** Select a check box to enable a particular folder to compile. Clear a check box to ignore that particular folder.
- **Status field:** The Status text field at the bottom of the display shows the current status of the Geographic Background Compiler.

2.2. Log Tab

The Log tab is where you can inspect log entries generated by the Geographic Background Compiler as you interact with the application.

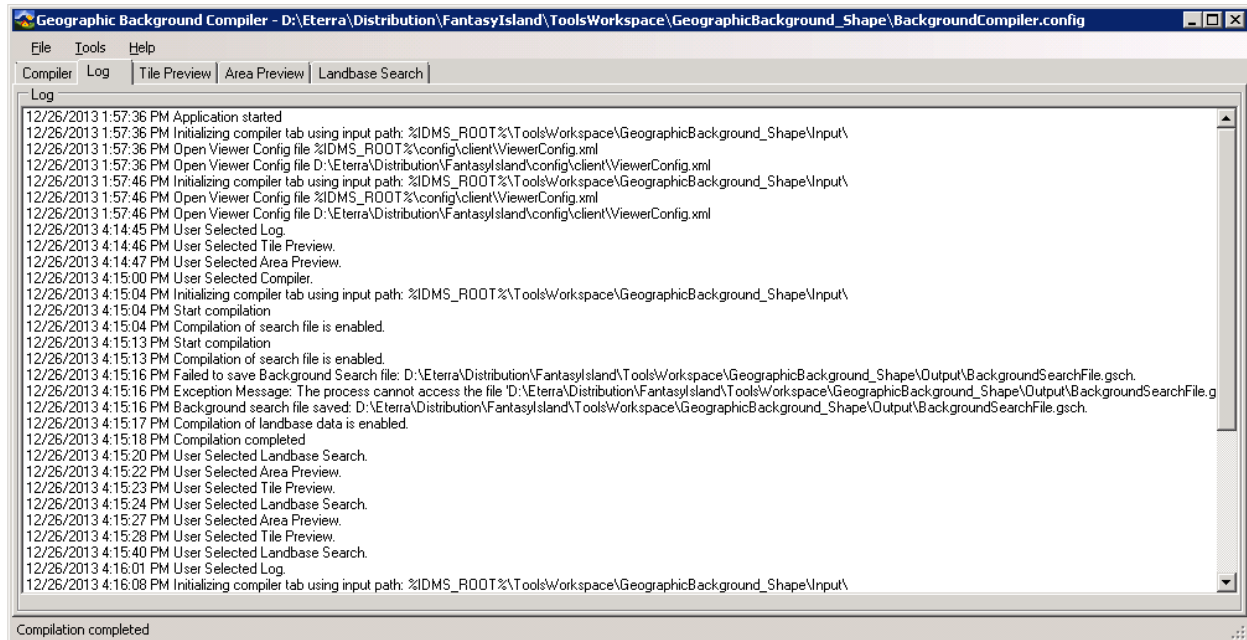


Figure 6. Geographic Background Compiler - Log Tab

When the Geographic Background Compiler is run from the command prompt, the log messages are written to standard output. The messages can be piped to a log file by adding `>MyLogFile.log` to the command.

2.3. Tile Preview Tab

The Tile Preview tab allows you to see a preview of each landbase tile.

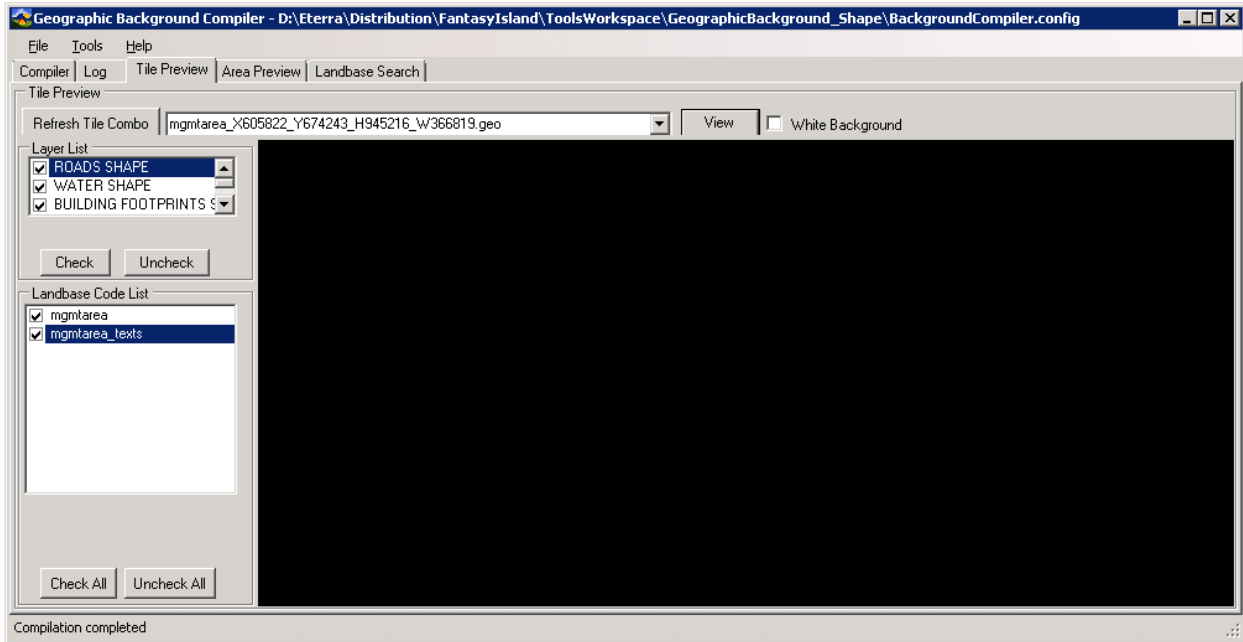


Figure 7. Geographic Background Compiler - Tile Preview Tab

The Tile Preview tab's fields and parameters include:

- **Refresh Tile Combo:** Click this button to populate the tile combo box with a compiled .GEO file from the output folder, and the drop-down list with all the compiled .GEO file names from the output folder.
- **View:** Click this button to display the tile preview.
- **White Background:** Select this check box to use a white background in the preview area. Clear this check box to use a black background.
- **Layer List:** Displays available layer names from the Viewer Configuration Editor. To populate the list, enter the correct path to the viewer configuration file via the Paths tab of the Configuration pop-up.
- **Landbase Code List:** The Landbase Code List shows the landbase codes of the selected tiles.

2.4. Area Preview Tab

The Area Preview tab can be used to see the "tile coverage".

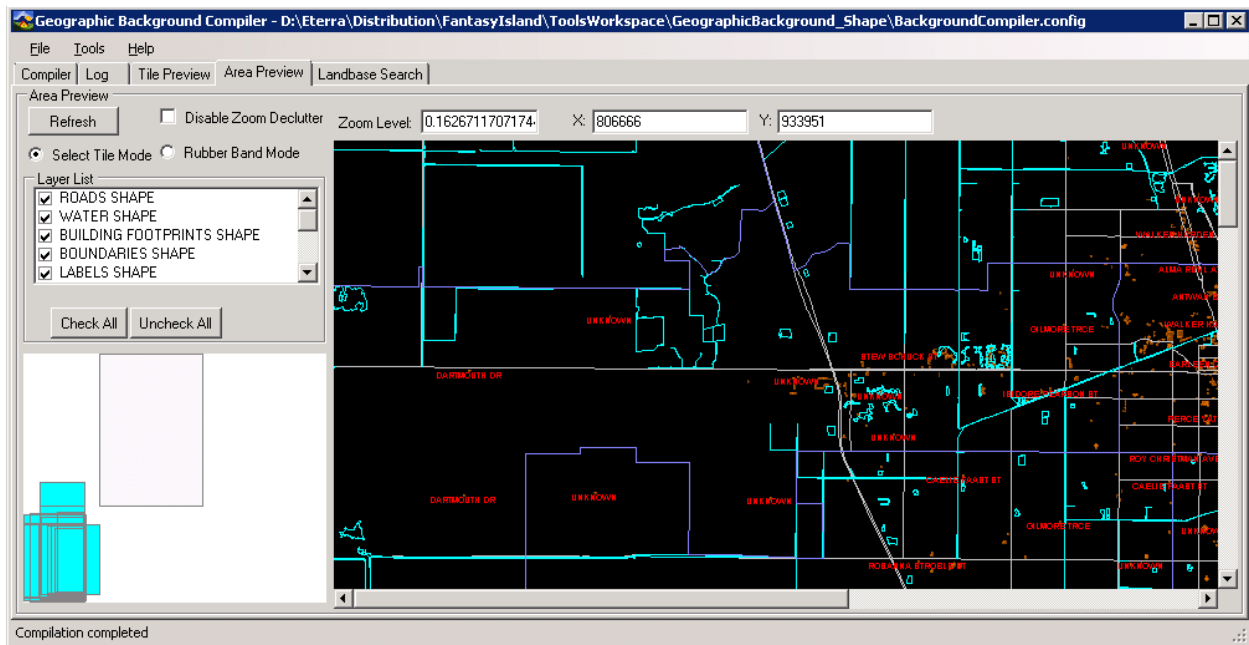


Figure 8. Geographic Background Compiler - Area Preview Tab

The Area Preview tab's fields and parameters include:

- **Refresh:** Click this button to preview a schematic of rectangles in the preview pane below the Layer List box.
- **Layer List:** Displays all the layer names from the Viewer Configuration Editor.
- **Disable Zoom Declutter:** Select to view lateral objects in the area preview pane.
- **Zoom Level:** Sets the zoom level applied to the area preview.
- **X:** Displays the horizontal coordinate value of the preview area.
- **Y:** Displays the vertical coordinate value of the preview area.
- **Select Tile Mode:** Select this option to preview the area by selecting a tile.
- **Rubber Band Mode:** Select this option to preview the area by the rubber-band selection. To use it, click on the rectangular preview pane, drag to the desired place and release the button. The area preview pane will show the area covered by the rubber-band selection.
- **Check All:** Click this button to display all the layers.
- **Uncheck All:** Click this button to hide all the layers.

2.5. Landbase Search Tab

The Landbase Search tab allows searching for landbase features and inspecting the compiled tiles generated by the Geographic Background Compiler without having to start the ADMS client.

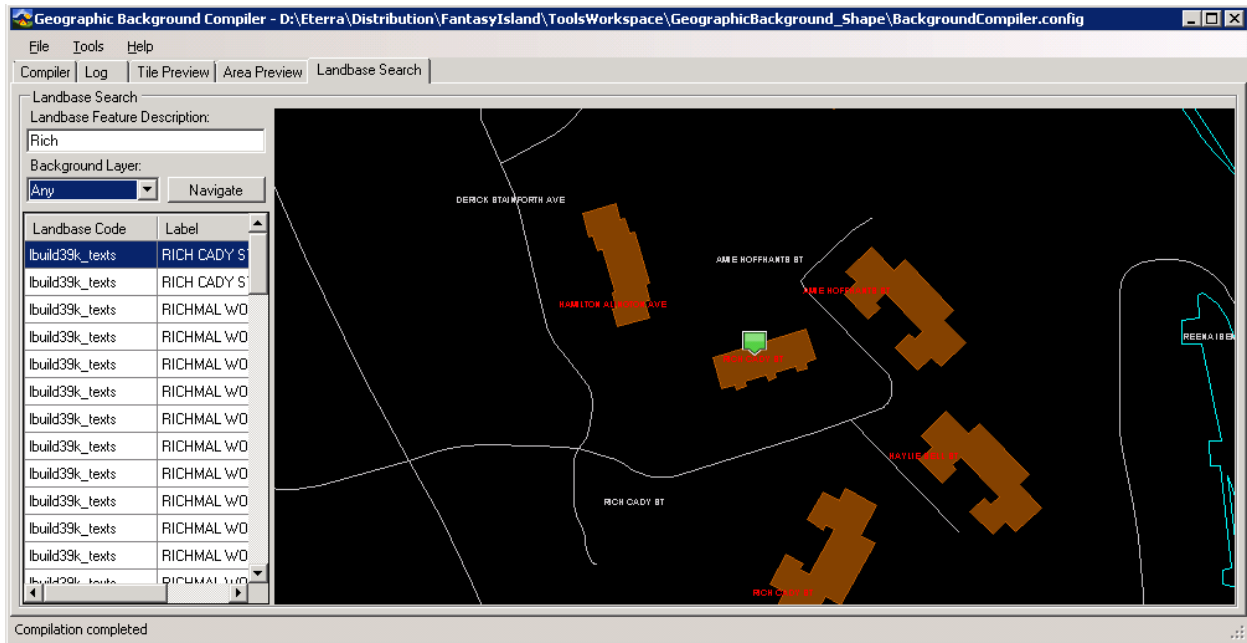


Figure 9. Geographic Background Compiler - Landbase Search Tab

The Landbase Search tab's fields and parameters include:

- **Landbase Feature Description:** Enter a name to view the search results.
- **Background Layer:** Select a background layer to be used for search and to be displayed on the landbase search pane.
- **Landbase Code:** Lists landbase codes of the labels.
- **Label:** Lists label names returned by the search query.

The Landbase Search is inactive and hidden by default and appears only when the background search file, `BackgroundSearchFile.gsch`, is compiled (via the Compile Background menu item on the File menu, or via folder compilation when the Compile Search check box is selected on the Options tab).

The Landbase Search tab contains the same search control as used on the Landbase Search page on the Tabbed Find Popup of the main ADMS user interface.

For a detailed description of the Landbase Search function, refer to the *Distribution Management System User Guide*.

2.6. Configuration Window

In the Configuration window you can configure the Geographic Background Compiler's settings and compile options.

To open the Configuration window, navigate to Tools > Configuration.

2.6.1. Options Tab

This section describes the settings provided on the Options tab of the Configuration window.

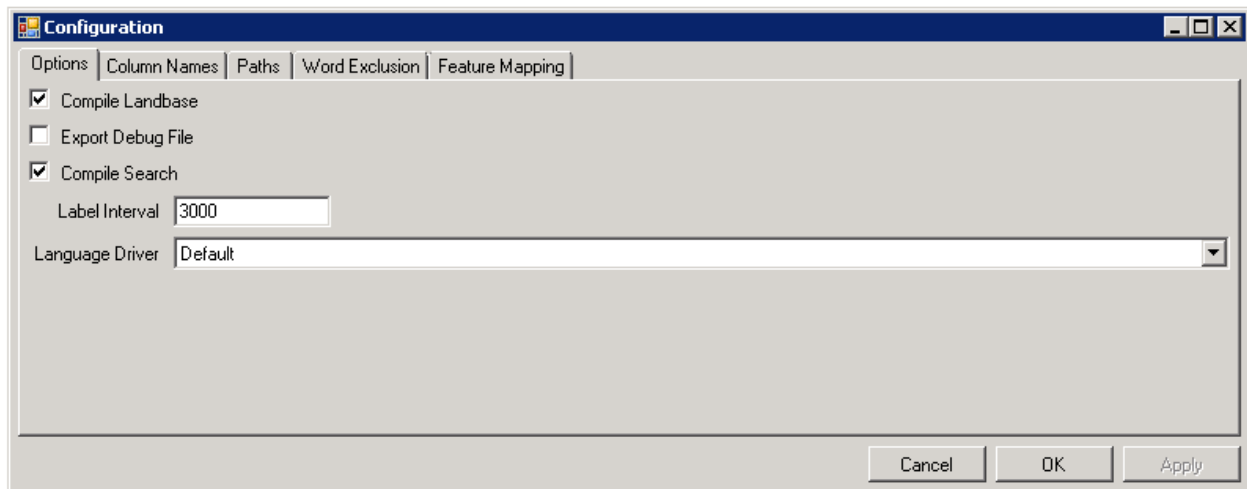


Figure 10. Configuration Window - Options Tab

- **Compile Landbase:** When selected, clicking Compile on the Compiler tab assembles landbase data from the input folder and saves it in the output folder.
- **Export Debug File:** If selected, a file is automatically created in the CSV format for each compiled landbase folder and is saved in the output folder. The CSV files have the same names as the corresponding GEO files and contain their rectangular coordinate values (MinX, MinY, MaxX, MaxY).
- **Compile Search:** When selected, clicking Compile on the Compiler tab automatically creates a BackgroundSearchFile.gsch file and saves it in the output folder.
- **Label Interval:** Sets the interval in local measurement units to repeat labels on the compiled background layer.

For example, if you want the Statesman label to be repeated along the road map every 5,000 feet, enter this value in the Label Interval field.

Note

Interval units correspond to the default geographic measurement units (for example, feet, meters, etc.). The default label interval is 3,000 units.

- **Language Driver:** Sets the language driver.

A language driver is required when special characters are used in background labels. For example, to correctly display French, Cyrillic, or Chinese symbols on the network view, the language driver must be set to UTF-8.

2.6.2. Column Names Tab

The Column Names tab allows you to specify the names of the table columns that the Geographic Background Compiler queries to compile the landbase information. These columns are usually contained in the *.dbf GIS files.

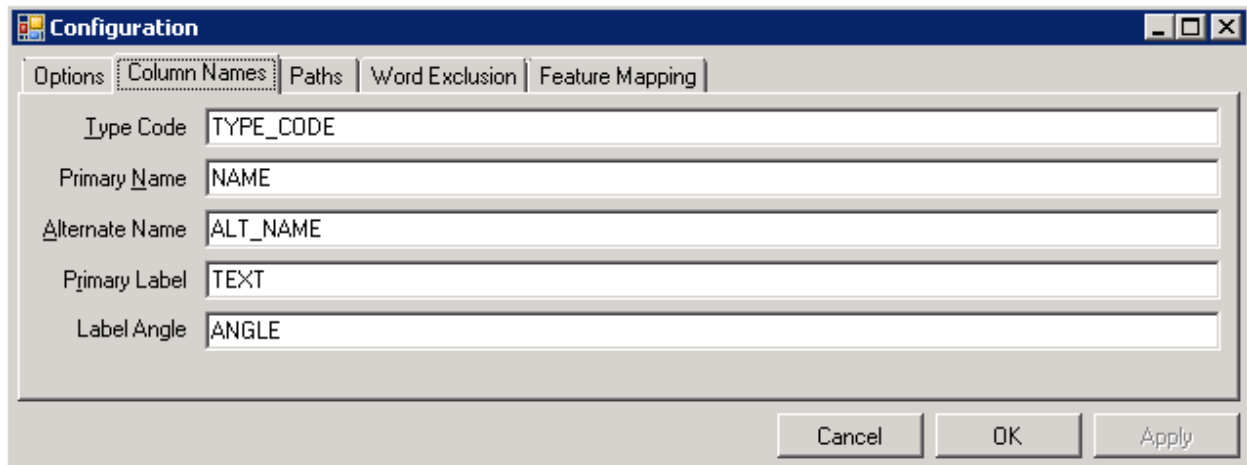


Figure 11. Configuration Window - Column Names Tab

- **Type Code:** For future use. Currently disabled.
- **Primary Name:** For future use. Currently disabled.
- **Alternate Name:** For future use. Currently disabled.
- **Primary Label:** The compiler queries the TEXT column of the GIS files for the labels.
- **Label Angle:** The compiler queries the ANGLE column of the GIS files for the angles that are applied to labels.

For correct compilation of the landbase data, the Column Name values must correspond to the names of the queried columns in the GIS files.

2.6.3. Paths Tab

The Paths tab defines paths to folders and files operated by the Geographic Background Compiler.

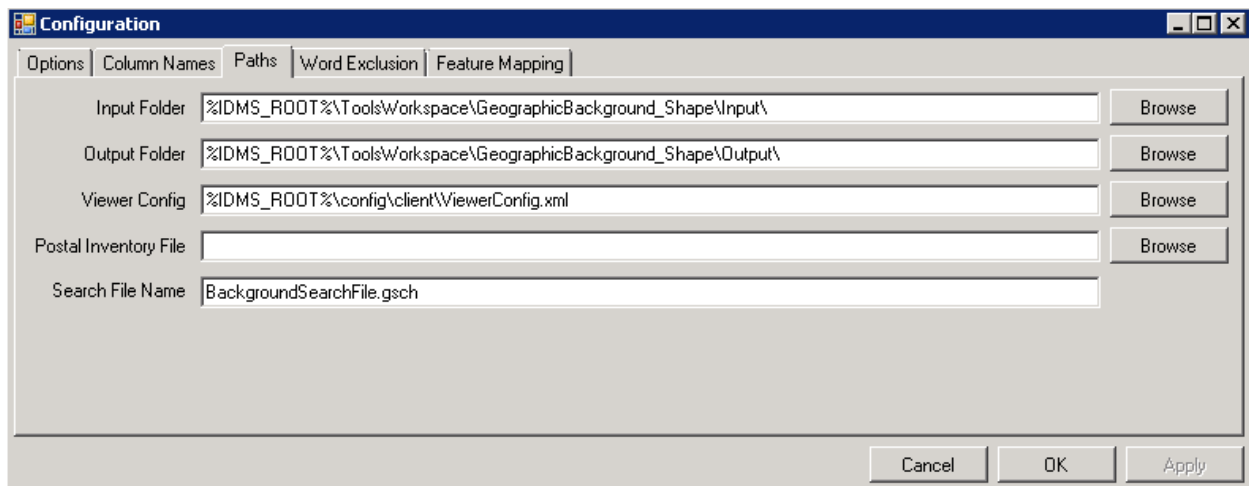


Figure 12. Configuration Window - Paths Tab

- **Input Folder:** Click Browse to set the path to the location of the exported files from GIS.
- **Output Folder:** Click Browse to set the path to the Compiler's output folder.
- **Viewer Config:** Click Browse to set the path to the viewer configuration file.
- **Postal Inventory File:** Click Browse to set the path to the Postal Inventory file used to map postal (ZIP) codes to towns.
- **Search File Name:** Sets the location and name of the background search file to be created during the compilation. When blank, the BackgroundSearchFile.gsch file will be produced in the output folder.

2.6.4. Word Exclusion Tab

The Word Exclusion tab allows you to optimize landbase searching by excluding certain words from search queries.

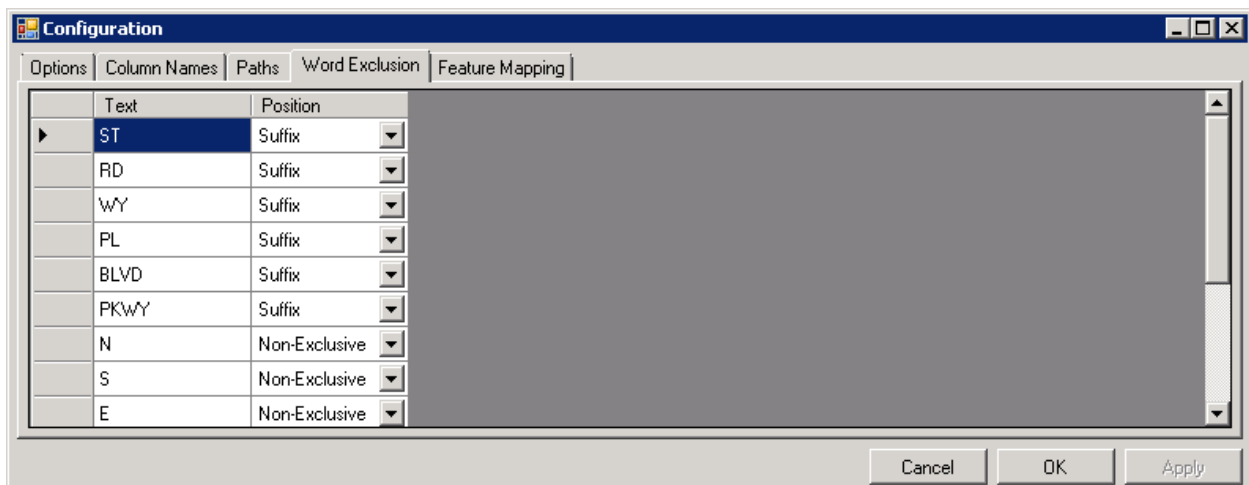


Figure 13. Configuration Window - Word Exclusion Tab

- **Text:** Contains the text to not search on the tool's Landbase Search tab and the Tabbed Find Popup of the ADMS client.
- **Position:** Sets the position of the text to be excluded from the search process.

If you select the Suffix position from the drop-down list, the search process excludes this text only if it appears at the end of the text label. Selecting Prefix excludes text that appears at the beginning of the text label. Selecting Any or Non-Exclusive excludes text in any position of the text label.

2.6.5. Feature Mapping Tab

The Feature Mapping tab allows you to map type codes from various shape files to the defined layers/levels.

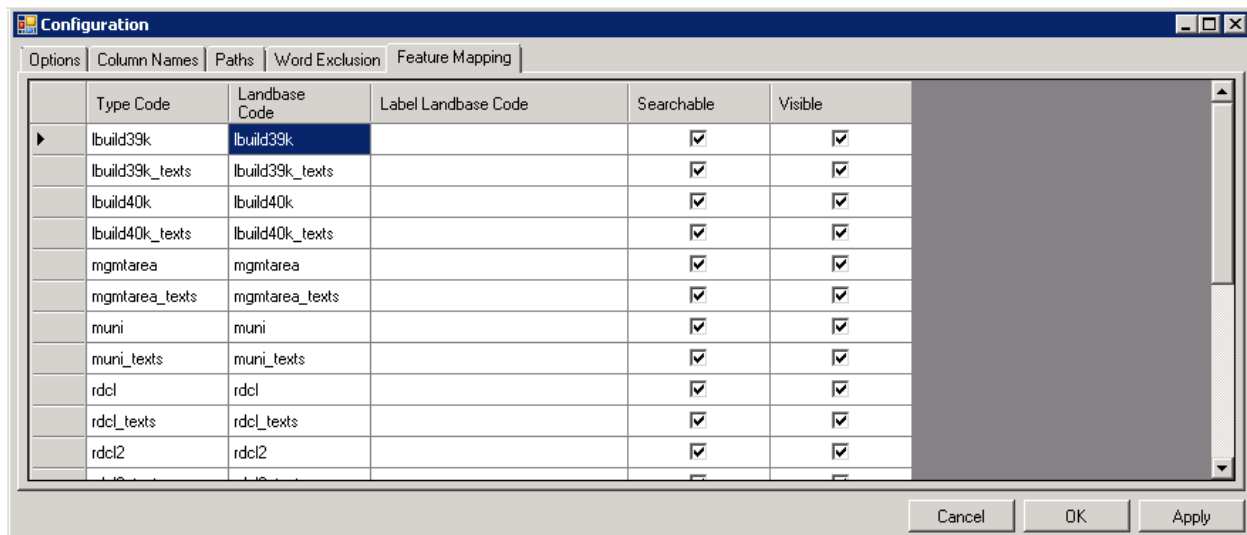


Figure 14. Configuration Window - Feature Mapping Tab

- **Type Code:** The code assigned to each landbase shape when it is created in the GIS. For example, a poly line with the type code value "rdcl" describes a certain type of road.
- **Landbase Code:** The string that can be used to set a human-readable name to an individual landbase feature. For example, the type code "rdcl" can be mapped to the landbase code "Highways".

The landbase code can also be used to map a number of type codes to one landbase code. For example, the type codes "rdcl", "rdcl1", "rdcl2", and "rdcl3" all describe different types of roads. For grouping purposes, they can all be mapped to the single landbase code "Roads".

Landbase codes function as an intermediate mapping between the "not so readable" type code values found in the individual shape files and the landbase "layers" presented to the end user in the ADMS client.

- **Label Landbase Code:** Sets the label landbase code to ungroup the layer from the landbase feature. This allows supporting the labels-only layer and enabling the label configuration

options (for example, zoom or show/hide) on the geographic network view.

- **Searchable:** Select to add the type code to the background search file (.gsch) and make it searchable on the geographic network view.
- **Visible:** Select to add the type code to the background search file (.gsch) and make it visible on the geographic network view. If unselected, the compiled type code will be invisible on the geographic network view.

3. General Usage

This chapter describes general functions of the Geographic Background Compiler.

3.1. Starting the Geographic Background Compiler

The Geographic Background Compiler can be launched from the Windows Start menu by navigating to All Programs > Eterra > Distribution > Tools > Modeling Tools > Geographic Background Compiler.

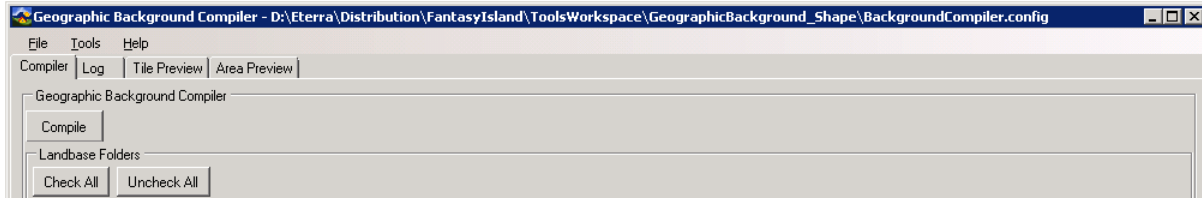
In the Geographic Background Compiler, the File menu includes the following commands:

- **New:** Creates a new Geographic Background Compiler configuration file.
- **Open:** Shows the Open dialog box to open an existing configuration file.
- **Save:** Saves the current configuration file.
- **Save As:** Saves the current configuration file with a different name.
- **Compile Background:** Compiles one or many landbase folders, and also creates the BackgroundSearchFile.gsch file when the Compile Search option on the Options tab is selected. The Compile Background option performs the same function as the Compile button on the Compiler tab.
- **Import > Type Codes:** Imports a Type Code file in Comma-Separated-Value (CSV) format and populates the Feature Mapping tab with the file's data.
- **Export > Excel Files:** Extracts the shape file data into a Microsoft Excel file and stores it in the output folder.
- **Export > Type Codes:** Exports the type codes that exist in different shape files and generates type code files in the output folder.
- **Export > Shape File Schemas:** Extracts the shape file data into an XML file and stores it in the output folder.
- **Exit:** Closes the application.

3.2. Loading a Geographic Background Compiler Configuration File

1. Start the Geographic Background Compiler.

The title bar of the Geographic Background Compiler indicates the configuration file being used, as shown below:



The default configuration file, BackgroundCompiler.config, is automatically loaded on startup.

2. Select File > Open, and choose an alternative configuration file.

The Geographic Background Compiler title bar changes to indicate the new file being used.

3.3. Saving a Geographic Background Compiler Configuration File

1. From the Tools menu, select Configuration to open the Configuration window.
2. Open the Paths tab and set paths to the input folder, output folder, and Viewer Configuration file location as needed.
3. Click OK to save changes and close the window.

3.4. Assigning Type Codes to Landbase Codes

1. From the Tools menu, select Configuration to open the Configuration window.
2. Open the Feature Mapping tab.
3. Enter the landbase code entries for each selected type code as needed.
4. Select the Searchable option to add the type code to the background search file (.gsch) and make the selected feature searchable on the geographic network view.
5. Select the Visible option to make the type code visible on the geographic network view.

If this option is not selected, the type code is compiled but becomes invisible on the ADMS client.

6. Click OK to save changes and close the window.

3.5. Compiling Landbase Folders

1. On the Compiler tab, under Landbase Folders, click the Check All button to compile all the landbase folders, or select individual landbase folders to compile only the selected folders.
2. Click the Compile button.

The Processing pop-up tracks the compilation process and shows the folders being compiled.

The Status field at the bottom of the display shows the current status of the Compiler. When finished, the Status text field displays the "Compilation Completed" message.

The Log tab contains compile messages for each folder and each compile completion.

The specified output folder contains the final compiled files, one for each selected folder.

3.6. Viewing Geographic Layers on the Tile Preview Tab

1. From the Tools menu, select Configuration to open the Configuration window.
2. Open the Paths tab and browse to the correct Viewer Configuration file location.
3. Click OK to save changes and close the Configuration window.
4. On the Tile Preview tab, click Refresh Tile Combo.

The combo box populates with a compiled .GEO file from the output folder location, and the combo box drop-down list contains all the compiled .GEO file names from the output folder location.

The Layer List box populates with all the layer names from the Viewer Configuration Editor, and the Landbase Codes list populates with all the landbase code names from the Viewer Configuration Editor.

5. To show a white background in the preview area, select the White Background check box, and then click the View button.

Alternatively, to show a black background, clear the White Background check box, and then click the View button.

6. To display a tile preview, select a file name from the combo box.

3.7. Viewing Geographic Layers on the Area Preview Tab

1. In the Geographic Background Compiler, open the Area Preview tab.
2. Click the Refresh button.

A schematic of rectangles is displayed in the preview pane below the Layer List box.

One major rectangle is displayed for each of the .GEO files in the specified output folder.

3. To display an area in the preview pane, select a rectangle.

When an area of overlap is selected, the subset of rectangles participating in the overlap is highlighted.

A different area is displayed for each highlighted rectangular area.

4. To "rubber-band" a selection of rectangles, click and drag in the rectangle area.

When multiple rectangles are selected, the corresponding tiles are displayed.

3.8. Performing Command-Line Operations

3.8.1. Command-Line Arguments

The Background Compiler provides a command-line interface, so that the compilation of landbase information can be incorporated into "build" scripts as part of the model management process.

Below is a list of command-line arguments that the Geographic Background Compiler supports.

-Batch

The argument used to flag that the program is running in a command-line background without the user interface.

-ConfigFile "*configuration file path*" or **-Config** "*configuration file path*"

The argument used to specify the location and name of the configuration file to use. This argument is required when running the Geographic Background Compiler from the command line. It can also be used in shortcuts to set the context when running the Geographic Background Compiler interactively.

-InputFolder "*folder path*" or **-Input** "*folder path*"

The argument used to specify the path to the input folder. This is an optional argument that, if presented, overrides the input folder configuration specified in the Geographic Background Compiler configuration file.

-OutputFolder "*folder path*" or **-Output** "*folder path*"

The argument used to specify the path to the output folder. This is an optional argument that, if presented, overrides the output folder configuration specified in the Geographic Background Compiler configuration file.

-Folder "*tile name*"

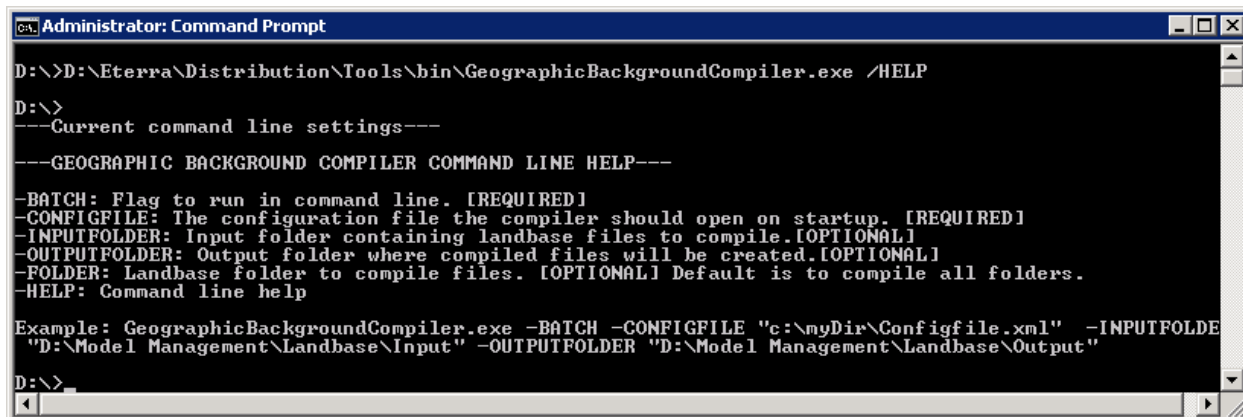
An optional argument used to specify the name of the landbase tile to compile. In this context, the tile name is equivalent to the name of the folder name describing the tile. If this argument is not provided, all tiles are compiled.

-bpath "*folder path*" or **-BitmapPath** "*folder path*"

The argument used to set a path to the location of the bitmap folder. The path must be entered exactly as defined in the Viewer Configuration Editor startup parameter `-bpath "bitmap path"`, where *bitmap path* is the full path to the folder.

-Help

The argument used to output descriptions of the available command-line arguments, as shown in the following image.



```
Administrator: Command Prompt
D:\>D:\Eterra\Distribution\Tools\bin\GeographicBackgroundCompiler.exe /HELP
D:\>
---Current command line settings---
---GEOGRAPHIC BACKGROUND COMPILER COMMAND LINE HELP---
-BATCH: Flag to run in command line. [REQUIRED]
-CONFIGFILE: The configuration file the compiler should open on startup. [REQUIRED]
-INPUTFOLDER: Input folder containing landbase files to compile.[OPTIONAL]
-OUTPUTFOLDER: Output folder where compiled files will be created.[OPTIONAL]
-FOLDER: Landbase folder to compile files. [OPTIONAL] Default is to compile all folders.
-HELP: Command line help
Example: GeographicBackgroundCompiler.exe -BATCH -CONFIGFILE "c:\myDir\Configfile.xml" -INPUTFOLDE
"D:\Model Management\Landbase\Input" -OUTPUTFOLDER "D:\Model Management\Landbase\Output"
D:\>
```

3.8.2. Starting the Geographic Background Compiler from the Command Prompt

1. Open the Windows command prompt.
2. Change to the ADMS bin directory.
3. At the command prompt, start the Geographic Background Compiler by typing:

```
> GeographicBackgroundCompiler /CONFIG "_configuration file path_"
```

In the case of a default system installation, the configuration file path may look as in the example below:

```
D:\Eterra\Distribution\Tools\bin\GeographicBackgroundCompiler.exe /CONFIG
D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\GeographicBackground_Shape\BackgroundCompiler.
config
```

For a list of optional command-line arguments to include in the startup command, refer to [Command-Line Arguments](#).

3.8.3. Compiling a Single Landbase Folder from the Command Prompt

1. Open the Windows command prompt.
2. At the command prompt, start the Geographic Background Compiler with the following arguments:
 - ConfigFile "specify the configuration file"
 - OutputFolder "specify the output folder"

- Folder "specify a single folder to compile"

The compiled file is located in the specified output folder.

Note

To create log messages containing the command-line switches used, the compilation of the single folder, and the successful completion of the compilation, add an argument to pipe the output to a file, as described in [Generating a Log File from a Command-Line Start](#).

3.8.4. Generating a Log File from a Command-Line Start

From the Windows command prompt, start the Geographic Background Compiler using the argument:

```
ConfigFile "specify the default configuration file" > "MyLogFile.log"
```

to generate a log file of all the Geographic Background Compiler debug messages.

The specified log file is created and contains the Geographic Background Compiler debug messages.

3.8.5. Creating and Running a Background Compiler Build Script

The example below demonstrates how to set variables and write commands to incorporate the compilation of landbase information into build scripts.

1. Create a script file (for example, D:\GEO.bat) and open it with a text editor.
2. Set a variable to point to the Geographic Background Compiler executable:

```
@echo off
SET GEOCOMPILER="D:\Eterra\Distribution\Tools\bin\GeographicBackgroundCompiler
.exe"
```

3. Set the variables:

```
SET
GeoConfigFile="D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\GeographicBackground_Shape\Back
groundCompiler.config"
SET
OutputPath="D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\GeographicBackground_Shape\Output"
SET
InputPath="D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\GeographicBackground_Shape\Input"
@echo on
DATE/T
TIME/T
```

4. Write a command.

The following is an example of a command that compiles one single landbase tile folder. This command is not normally used in a system build script, but it might be useful in some situations.

```
%GEOCOMPILER% -ConfigFile %GeoConfigFile% -Input %InputPath% -Output %OutputPath% -Folder "CNTYLINE"
```

The following is an example of a command that compiles all the landbase tile folders. This command is normally included in a system build script.

```
%GEOCOMPILER% -ConfigFile %GeoConfigFile% -Input %InputPath% -Output %OutputPath%
```

5. Open the script file.

6. Open the Windows command prompt.

7. At the command prompt, run the build script. Include an argument to capture the compiler's output messages in a dedicated log file, as shown below:

```
D:\>GEO.bat > Landbase_Build_log.txt
```

The log file captures all the output messages and documents the compiling process, as shown below:

```

LandBase_Build_log - Notepad
File Edit Format View Help
Setting env var: DMS_CLIENT_NETCONF=%IDMS_ROOT%\config\client\netconf_localhost.xml
Setting env var: DMS_RESEDIT_CONFIG=%IDMS_ROOT%\Toolsworkspace\ResourceEditor\ResourceEditorConfig.xml
Setting env var: DMS_SSE_CONFIG=%IDMS_ROOT%\Toolsworkspace\SubstationEditor\SubstationEditorConfig.xml
Setting env var: DMS_VIEWER_CONFIG=%IDMS_ROOT%\config\client\ViewerConfig.xml
Setting env var: DMS_CLIENT_CONFIG=%IDMS_ROOT%\config\client\OmsClientConfig.xml
Setting env var: DMS_SERVER_CONFIG=%IDMS_ROOT%\config\server\OmsServerConfig.xml
Setting env var: DMS_SNAPVIEWER_CONFIG=%IDMS_ROOT%\Toolsworkspace\SnapshotViewerConfig.xml
Setting env var: DMS_SWOCLIENT_CONFIG=%IDMS_ROOT%\config\client\SwitchOrderClientConfig.xml
Setting env var: DMS_DNOMDBLDR_CONFIG=%IDMS_ROOT%\Toolsworkspace\DNOMDatabaseLoader\DNOMDBLoaderConfig.xml
Setting env var: DMS_MMSSERVICE_ADRS=net.tcp://localhost:8085/ModelManagerService
Setting env var: DMS_CODEDTRANS_CONFIG=%IDMS_ROOT%\config\shared\CodedTranslationConfig.xml
Setting env var: DMS_BITMAPS=%IDMS_ROOT%\bitmaps
Setting env var: DMSMAIN_CONFIG=%IDMS_ROOT%\config\server\NetServerMainConfig_localhost.txt
Setting env var: DMSDNAC_CONFIG=%IDMS_ROOT%\config\server\NetServerDnacConfig_localhost.txt
Setting env var: DMSCOMBO_CONFIG=%IDMS_ROOT%\config\server\NetServerComboConfig_localhost.txt
Setting env var: DMSSIM_CONFIG=%IDMS_ROOT%\config\server\NetServerSimConfig_localhost.txt
Setting env var: DMSDOTS_CONFIG=%IDMS_ROOT%\config\server\NetServerSimConfig_localhost.txt
Setting env var: DMSRO_CONFIG=%IDMS_ROOT%\config\server\NetServerReadonlyConfig_localhost.txt
Setting env var: DMS_OMSCALL_CONFIG=%IDMS_ROOT%\config\server\CallServerConfig.xml
Setting env var: DMS_OMSRELAY_CONFIG=%IDMS_ROOT%\config\server\RelayServerConfig.xml
Setting env var: DMS_DATAREPLIC_CONFIG=%IDMS_ROOT%\config\server\NetDataReplicatorConfig.xml
Setting env var: DMSOMSCREW_CONFIG=%IDMS_ROOT%\config\server\CrewServerConfig.xml
Setting env var: DMSDATASRV_CONFIG=%IDMS_ROOT%\config\server\NetDataServerConfig_localhost.txt
Setting env var: DMS_SOCONFIG=%IDMS_ROOT%\config\server\SwitchOrderServerConfig.txt
Setting env var: DMS_CREWSIM_CONFIG=%IDMS_ROOT%\config\server\CrewsimulatorConfig.xml
Setting env var: DMS_ROBODISP_NETCFG=%IDMS_ROOT%\config\server\NetconfRoboDispatch.xml
Setting env var: DMS_ROBODISP_CONFIG=%IDMS_ROOT%\config\server\RoboDispatchConfig.txt
Setting env var: NETARCHIVE_CONFIG=%IDMS_ROOT%\config\server\ArchiveAppConfig.xml
Setting env var: DMSSTDY_CONFIG=%IDMS_ROOT%\config\client\NetServerStudyConfig.txt
Setting env var: DMS_CALLENTRY_CONFIG=%IDMS_ROOT%\config\client\CallEntryConfig.xml
Setting env var: DMS_EVENTSCR_NETCONF=%IDMS_ROOT%\config\client\netconfes.xml
Setting env var: DMS_DATASRV_CONFIG=%IDMS_ROOT%\config\client\netconf_dataservice.xml
Setting env var: DMS_INTERFACE_SRV_CONFIG=%IDMS_ROOT%\config\client\DmsInterfaceServerConfig.xml
---Current command line settings---

Start compilation
Compilation of search file is enabled.
Background search file saved: D:\Eterra\Distribution\FantasyIsland\Toolsworkspace\GeographicBackground
Compilation of landbase data is enabled.
Compilation completed

```

3.9. Changing the Default Bitmap Folder Location

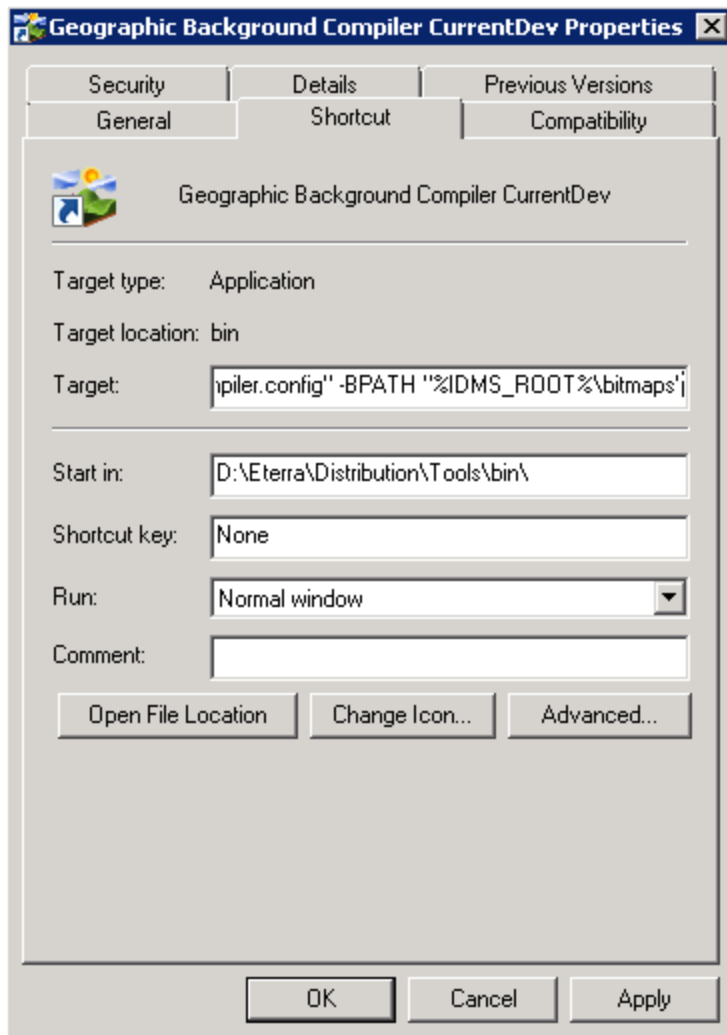
To change the default bitmap folder location for the Geographic Background Compiler:

Note

The path must be entered exactly as defined in the Viewer Configuration Editor's startup parameter (-bpath "*bitmap path*"), where *bitmap path* is the full path to the folder.

1. Locate the Geographic Background Compiler shortcut.
2. Right-click the application icon, and select the Properties option from the right-click menu.

The Geographic Background Compiler CurrentDev Properties window appears.



3. Select the Shortcut tab.
4. In the Target field, after the -configfile argument, locate the -bpath "*bitmap path*" argument, where *bitmap path* is the full path to the folder. Enter the new path exactly as defined in the Viewer Configuration Editor.
5. Click Apply.
6. Click OK.

The Geographic Background Compiler looks for a bitmap folder using the new path.

4. Applying the New Background Data

This chapter describes how to generate new background data and apply it to the ADMS client.

Updating the geographic background data requires three separate processes:

- Generating layers and the search file in the Geographic Background Compiler.
- Defining background parameters in the Net Dynamics Configuration Editor.
- Adding and configuring layers in the Viewer Configuration Editor.

The following sections provide information about how to generate shape files, import type codes from shape files, create a dedicated background layer with a human-readable name, update the client configuration, associate the background layer with a filter button, and add the layer to the toolbar on the geographic network view.

4.1. Generating Background Layers

To configure the tool and generate shape tiles used by the Distribution Management System client:

1. Start the Geographic Background Compiler.

The Background Compiler starts, and the default configuration file `BackgroundCompiler.config` is automatically loaded.

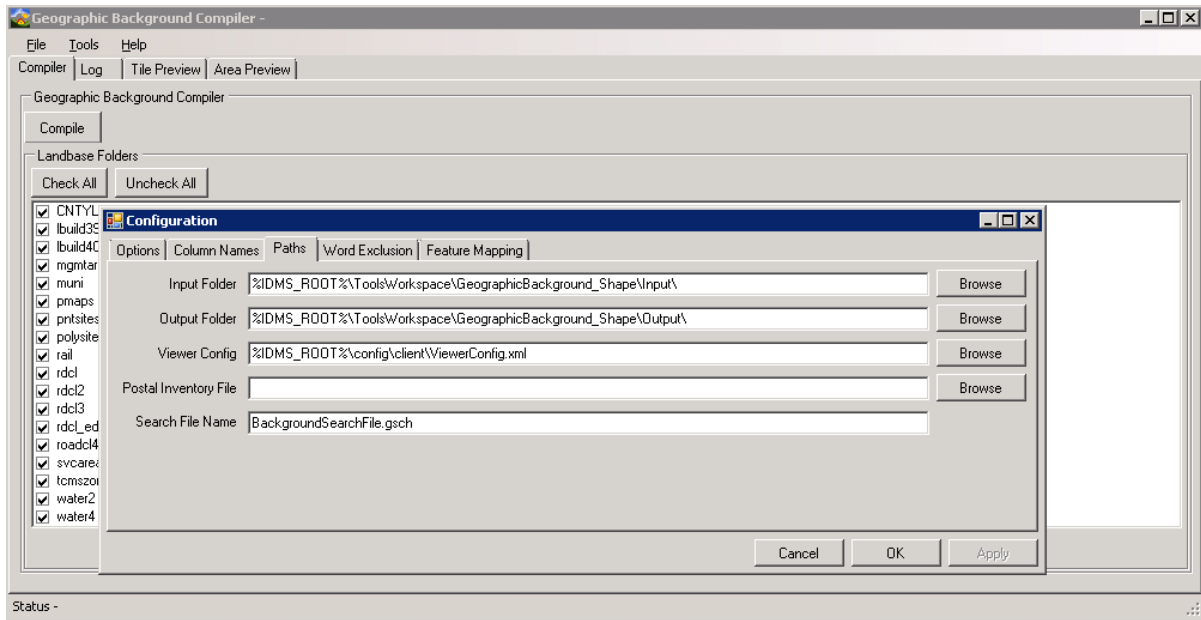
2. From the Tools menu, select Configuration to open the Configuration window.

3. Open the Paths tab and set the following parameters as needed:

- Full path to the input folder where the Compiler accesses the files exported from the GIS
- Full path to the output folder where the Compiler writes out the background files produced
- Full path to the Viewer Configuration file to support a realistic preview option on the Tile Preview tab

The Tile Preview feature draws the content of the compiled tiles according to line and text colors specified in the configuration file.

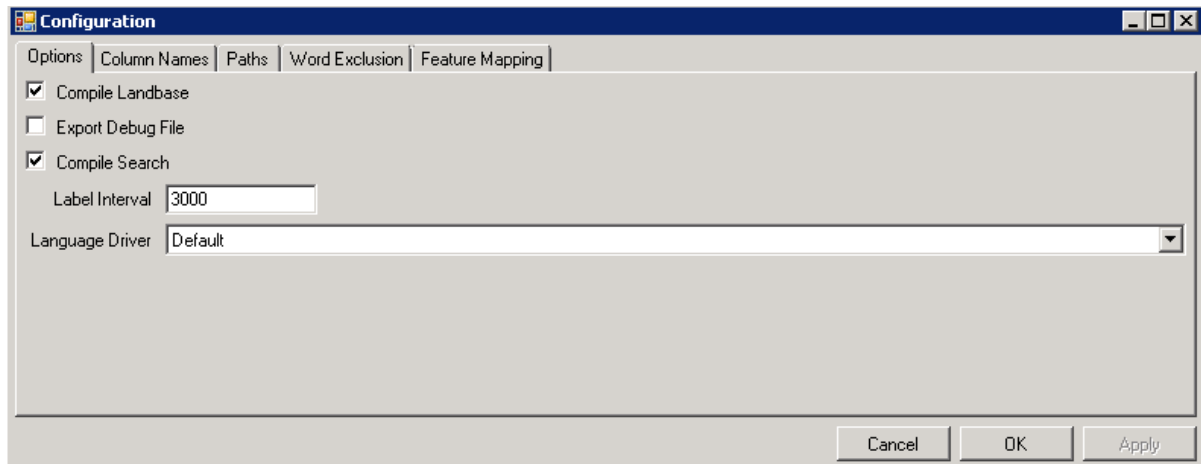
- Full path to the Postal Inventory file
- Name and path of the background search file to be produced by the Compiler



4. Open the Options tab and set the following parameters as needed:

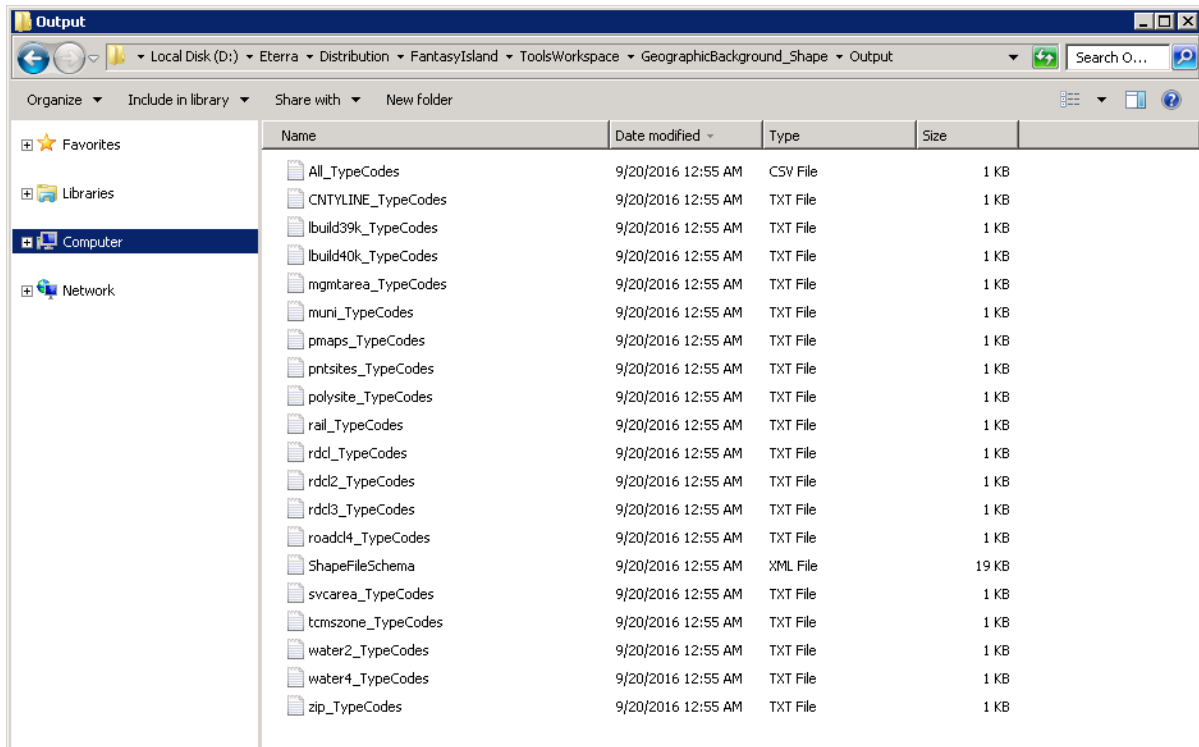
- Options to compile the landbase and search files
- Option to export debug files
- Interval in which labels will be displayed on the compiled background layer
- Language Driver from the drop-down list

The example below illustrates how these parameters can be entered.



5. From the Geographic Background Compiler's File menu, select Export > Type Codes.

The output folder is populated with the type codes files, as shown in the example below:



6. From the File menu, select Import > Type Code.

7. Navigate to the output folder and select the All_TypeCodes.csv file.

8. Click the Open button.

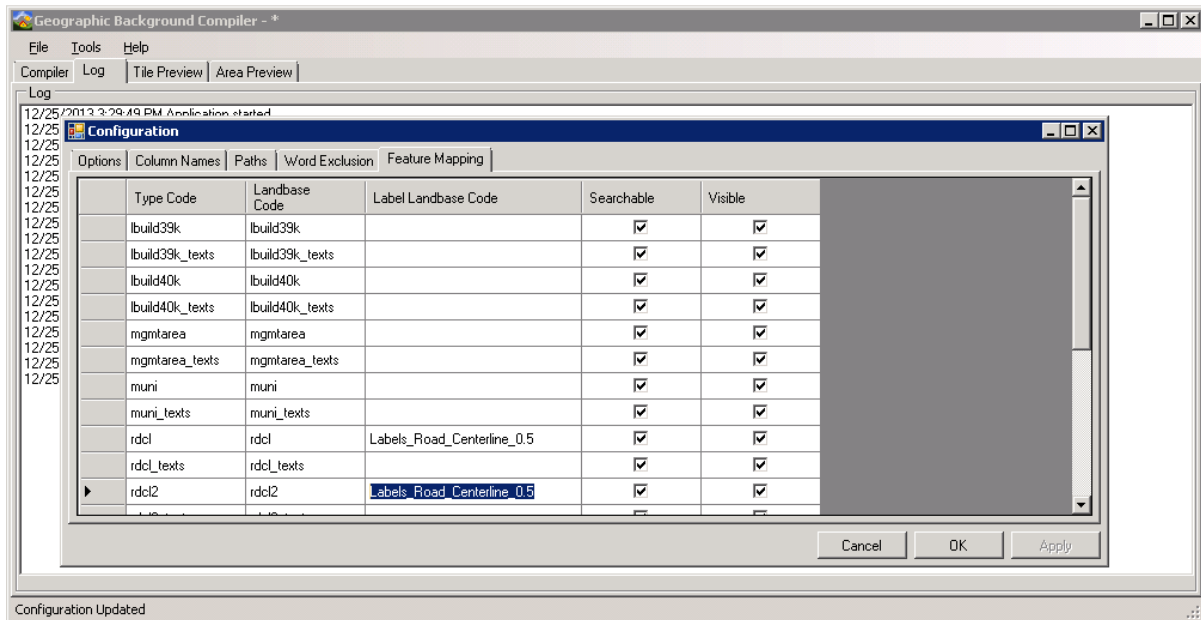
9. From the Tools menu, select Configuration to open the Configuration window and navigate to the Feature Mapping tab.

The Feature Mapping tab is populated with type codes.

10. On the Feature Mapping tab, enter and modify the following options:

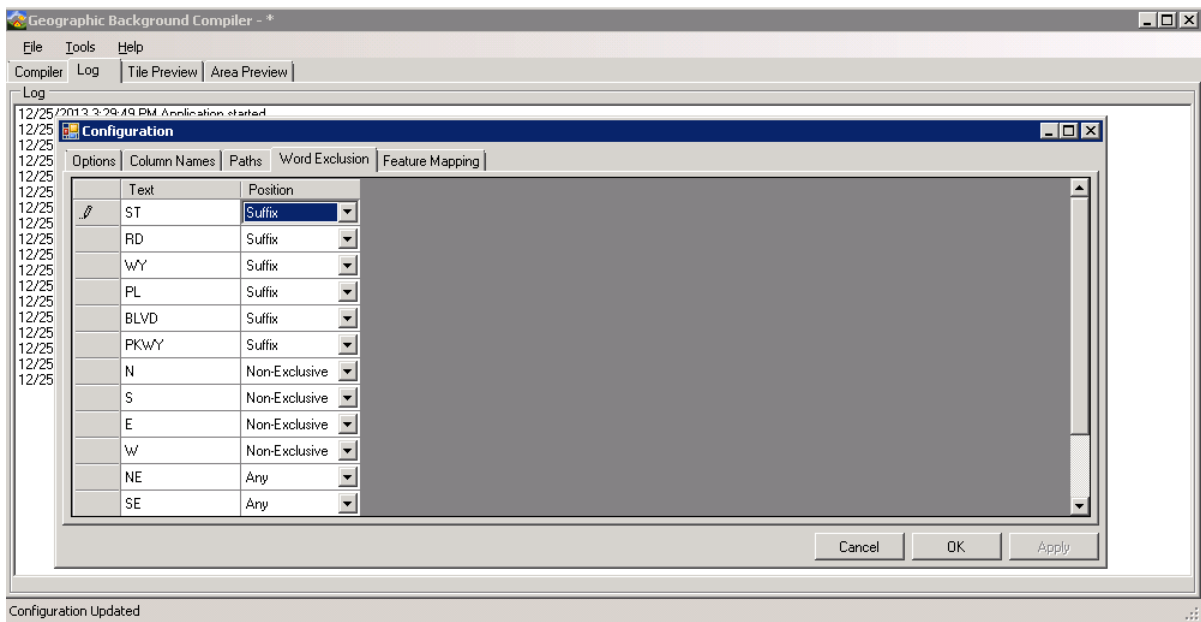
- Enter the landbase code
- Enter the label landbase code
- Select the Searchable option to make the landbase feature searchable on the geographic network view. The flag for the selected type code is added to the background search file.
- Select the Visible option to make the selected type code visible on the geographic network view. If the flag is not set, the type code is compiled but not visible on the ADMS client.

In the figure below, all the type codes have been mapped to landbase codes. In this case, no grouping of landbase features has been used. Normally, a number of similar type codes are mapped to a single landbase code that describes a landbase feature.



While the Geographic Background Compiler processes individual tiles during compilation, all landbase features are grouped according to the landbase code specified on the Feature Mapping tab. If a given type code does not exist on the Feature Mapping tab, all the shapes with that type code are excluded. This gives you the option to remove (or "compile out") the landbase features not needed at run time.

11. On the Word Exclusion tab of the Configuration window, set the text to non-searchable by the landbase search as appropriate.



For example, to make all suffixes "ST" non-searchable, enter "ST" into the Text field and select Suffix from the drop-down list in the Position field.

To add a new exclusion, scroll to the end of the list and add a new entry in the empty line marked with an asterisk.

The exclusions will be recorded when the background search file is compiled.

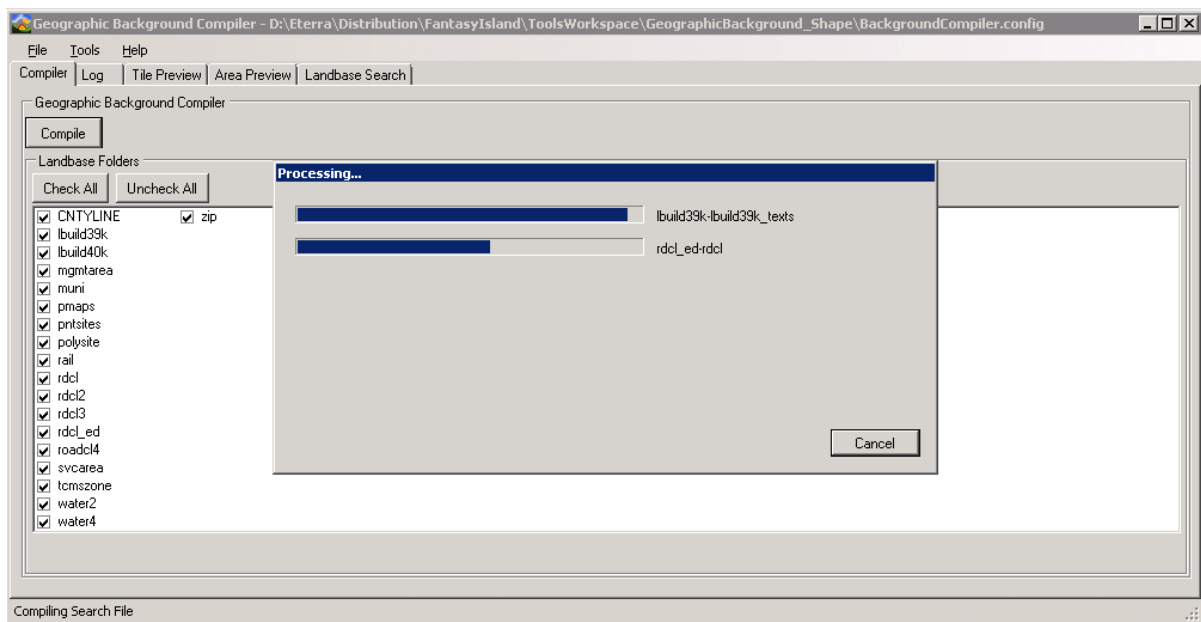
12. On the Geographic Background Compiler, select the Compiler tab.

The Compiler tab appears, displaying a list of all folders found in the input folder.

13. Select one or more folders to compile by selecting a check box, or click the Check All button to select all the folders.

14. To start the compilation of the selected folders, click the Compile button.

The Compiler reports progress in the compilation pop-up, as shown below, and logs information on the Log tab. The Status text field displays the current status of the Compiler.



Once the compilation is complete, the data is saved in the .geo files stored in the output folder. In addition, the background search file is created.

4.2. Defining Background Parameters in the Net Dynamics Configuration Editor

The ADMS client uses background shape files to show background information. It requires the search file to operate the landbase search.

The parameters that define displaying the background data are configured using the Net Dynamics Configuration Editor.

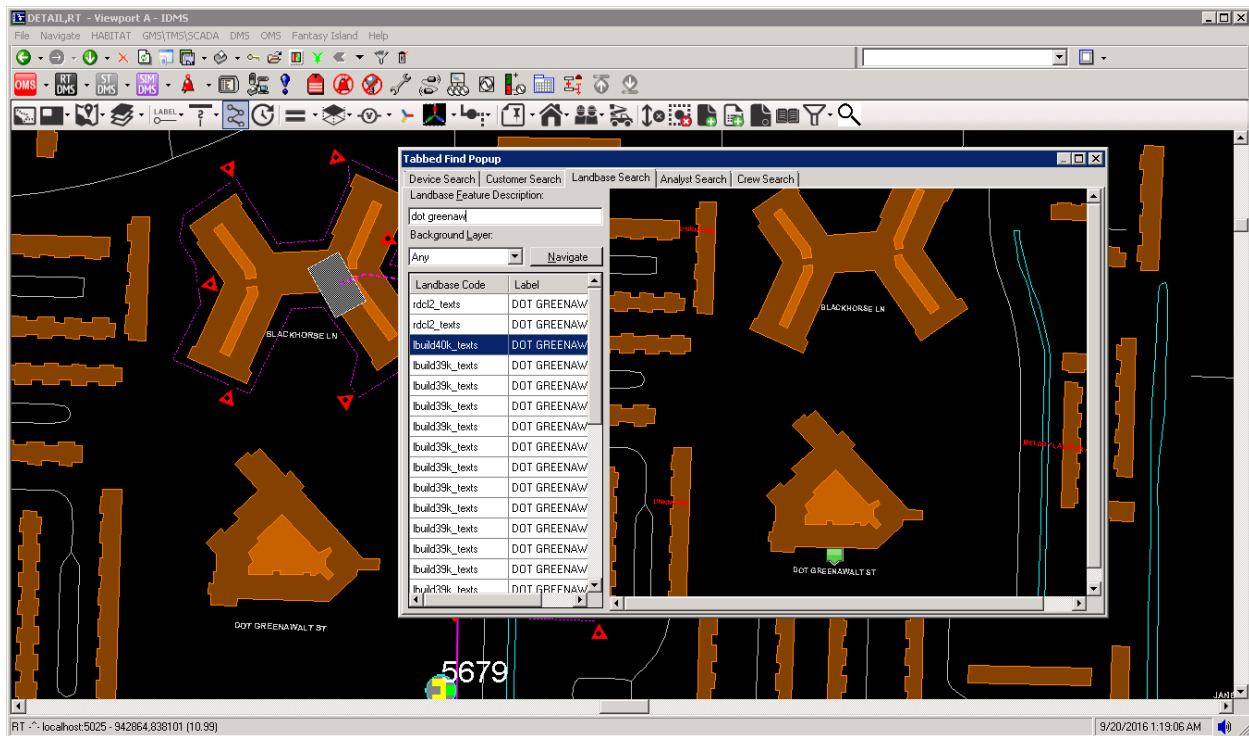


Figure 15. Finding Background Data in the Client

To specify the background parameters:

1. Open the Net Dynamics Configuration Editor.

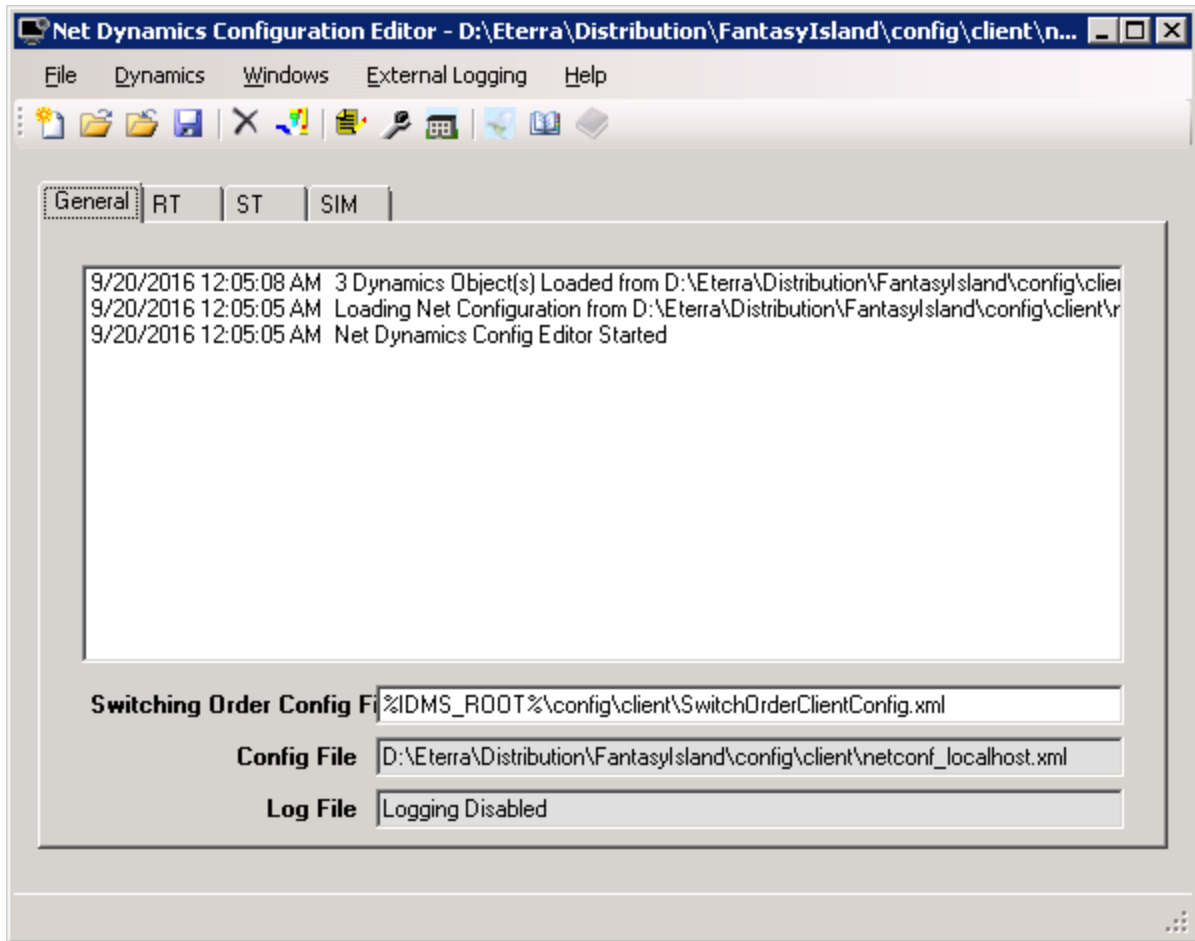
The default installation path is:

D:\Eterra\Distribution\Tools\bin\NetDynamicsConfigEditor.exe

The Net Dynamics Configuration Editor starts.

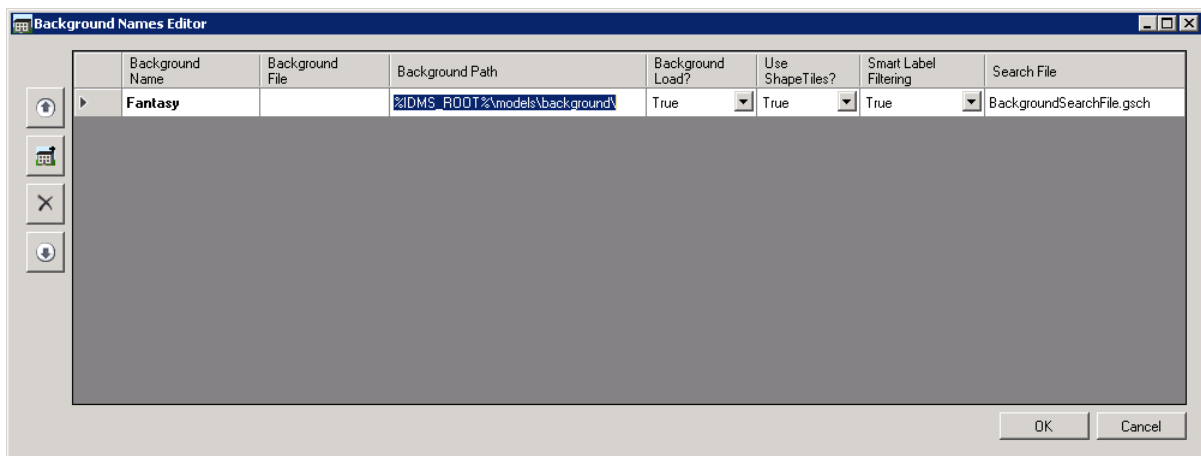
2. Select File > Open, and select the netconf_localhost.xml file from the list on the Open pop-up display, where localhost is a server name.

The network configuration is uploaded, and three new tabs named "RT" (real-time), "ST" (study), and "SIM" (training simulator) appear, as shown below.



3. Click the Backgrounds Editor button .

The Background Names Editor window is displayed.



4. In the Background Path field, specify the path to the folder that stores shape files and the search files.
5. In the Search File field, specify the name of the search file.

6. Copy your generated shape files and the search file to the specified Background folder.
7. Select the True option for the Use Shape Tiles.
8. Click OK.
9. Save the changes via File > Save.

For more information about other configuration parameters, refer to the *Net Dynamics Configuration Editor User Guide*.

4.3. Defining a New Background Layer in the Viewer Configuration Editor

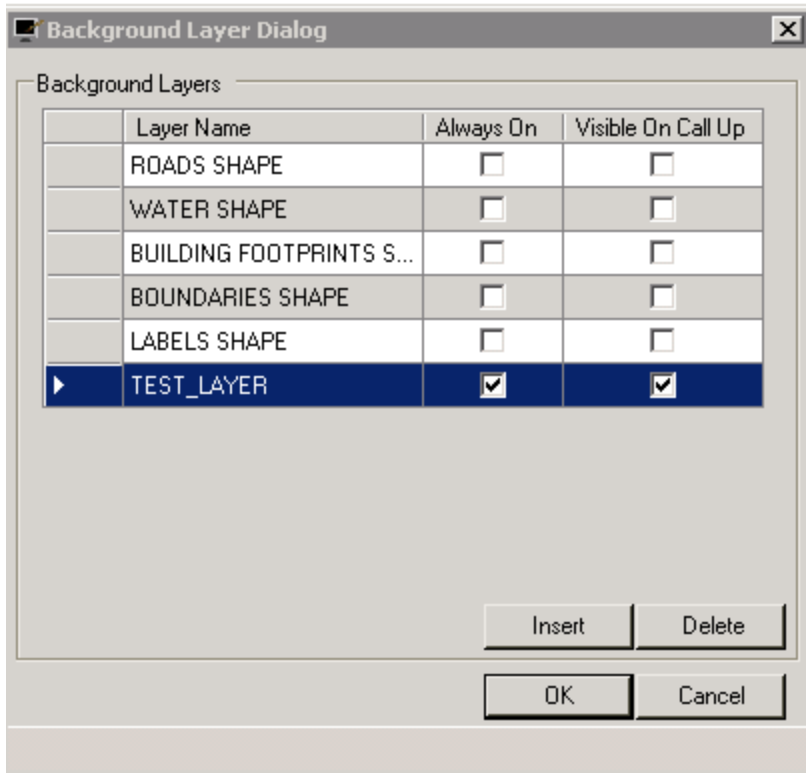
You can add and configure a new layer that has been defined in the Geographic Background Compiler using the Viewer Configuration Editor.

To add a new layer:

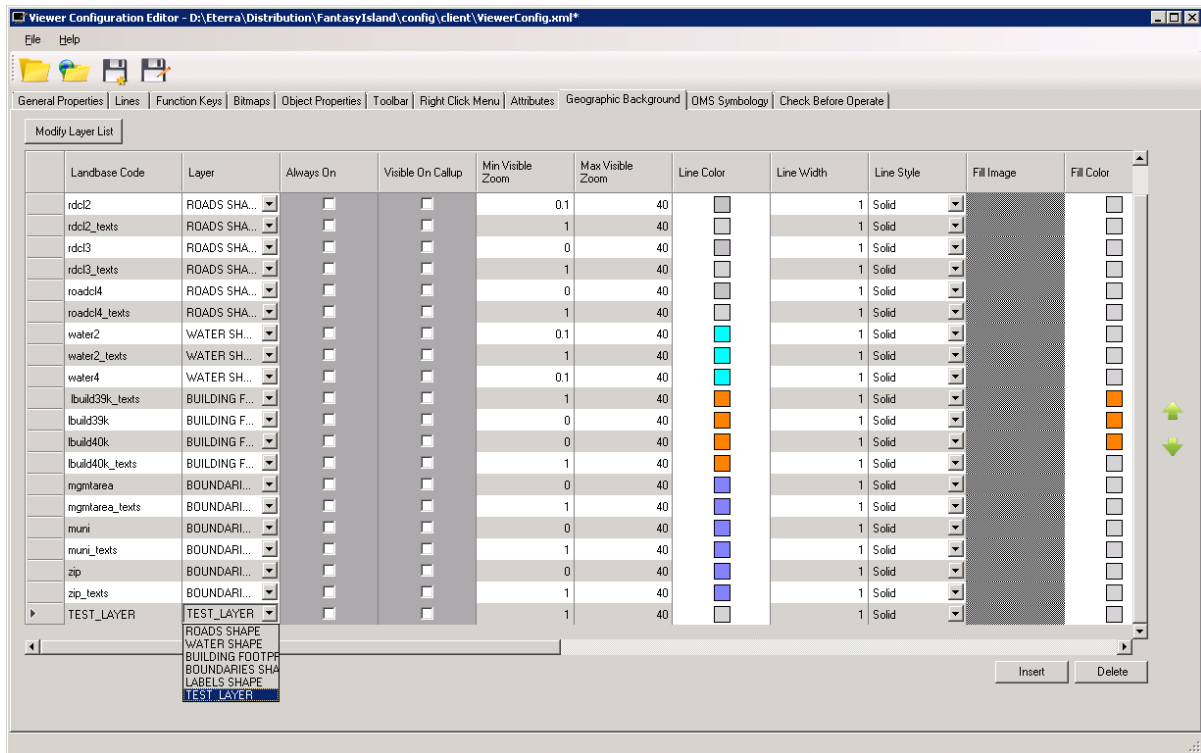
1. Open the Viewer Configuration Editor.
2. Select the Geographic Background tab of the Viewer Configuration Editor.
3. Click the Modify Layer List button at the top of the display.

The Background Layer dialog appears.

4. Scroll to the end of the Background Layers listing.
5. To add a new entry, click the Insert button.



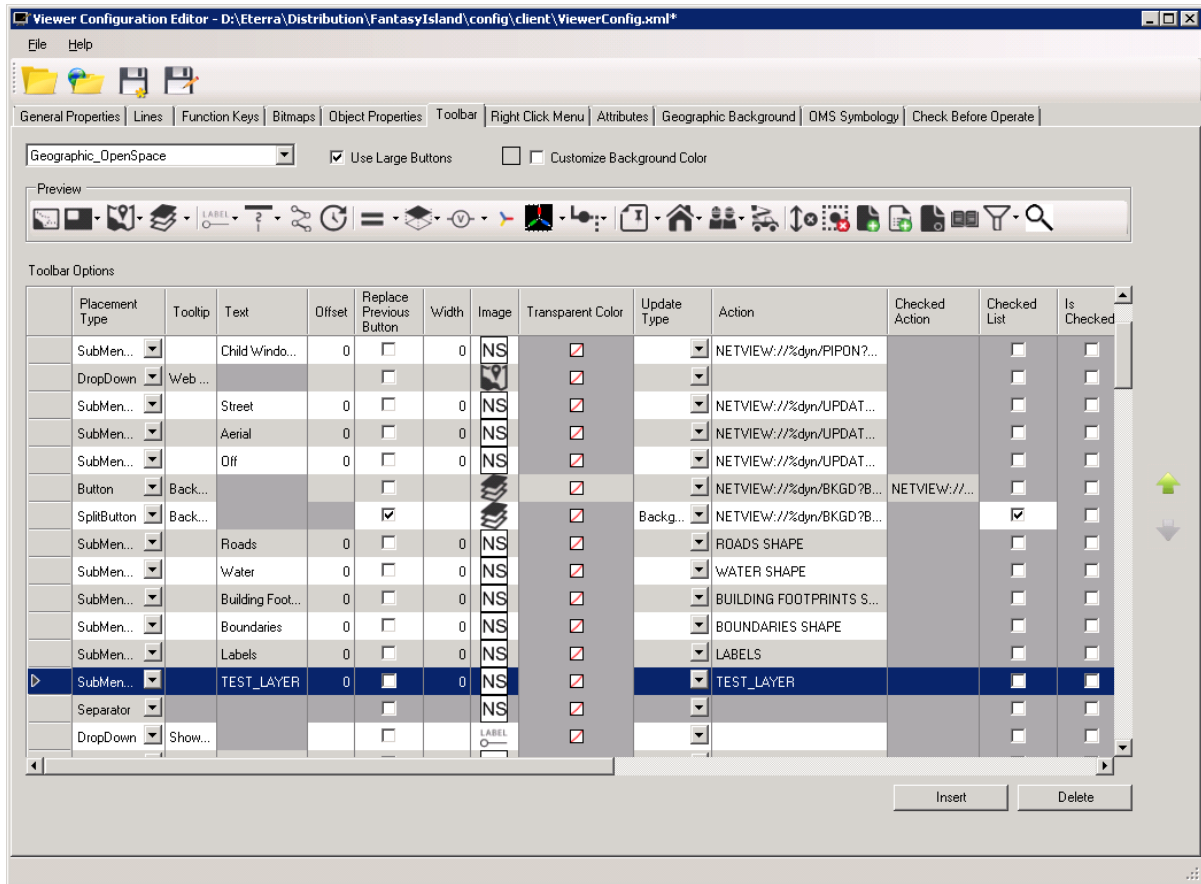
6. Enter the name of the new background layer, as shown above.
7. Select the following check boxes:
 - **Always On:** When selected, this layer will always appear on the geographic display of the ADMS client. It cannot be hidden from view.
 - **Visible On Call-Up:** When selected, this layer will appear on the display when initially called up. You can later choose to turn this layer off.
8. Click OK.
9. On the Geographic Background tab, select the Layer drop-down list to verify that it contains the new entry, as shown below:



4.4. Adding a New Background Layer Label to the Geographic_OpenSpace (Common) Toolbar

To add a label for a geographic background layer to a Geographic_OpenSpace toolbar:

1. Open the Viewer Configuration Editor and navigate to the Toolbar tab.
2. Select the Geographic_OpenSpace from the toolbar drop-down list.
3. Scroll down to find the existing Background Layers section.
4. To add a new entry, click the Insert button.
5. For a new background layer, enter the following information (leave the remaining fields blank):
 - Offset = 0
 - Width = 0
 - Image = Bitmap file name for the layer
 - Text = Background layer name
 - Action = Background layer name
 - Is Checked = False
6. Use the Up and Down arrow buttons at the right of the Toolbar Options grid to move the new entry to the location of the Background Layers button group.



7. To save the changes to the viewerconfig.xml file, click Save from the File menu.

4.5. Adding a New Landbase Code to the Viewer Configuration Editor

You can add a new landbase feature to the Geographic Background tab of the Viewer Configuration Editor and map it with a layer using the following steps:

1. Open the Viewer Configuration Editor.
2. Select the Geographic Background tab of the Viewer Configuration Editor.
3. To add a new entry, click the Insert button.
4. Enter the new landbase code exactly as it appears on the Feature Mapping tab of the Geographic Background Compiler.
5. From the Layer drop-down list, select a layer that you want to associate with the new landbase code.
6. Configure other geographic background parameters, as described in the Geographic Background chapter of the *Viewer Configuration Editor User Guide*.

From the Viewer Configuration Editor, you can specify several visualization preferences (line color, max/min zoom, etc.) for each landbase code. For example, all types of roads such as interstates, highways, and minor roads can all be assigned to the Road layer. When you turn on the Road layer, all roads extracted from the landbase shape files are available to be drawn by the ADMS client. Each type of road is drawn according to the properties configured for the landbase code the road belongs to.

5. Modeling Structures

In the ADMS system, any physical structure that is not part of the electrical network model (DNOM) might be referred to as a structure object.

A structure object can be one of the following objects:

- **Point:** Refers to a single point object. For example, pole, streetlight, stop light, or radio-frequency mesh network device.
- **MultiPoint:** Refers to a collection of point objects. For example, network devices group A or callboxes.
- **LineString:** Refers to a segmented line object. For example, work boundary line or jurisdiction boundary line.
- **MultiLineString:** Refers to a collection of segmented line objects. For example, hazard area boundary lines.
- **Polygon:** Refers to a polygon object. For example, restricted area perimeter.
- **MultiPolygon:** Refers to a collection of polygon objects. For example, fire sensitive area perimeters.
- **Point (Text Label):** Refers to a text label. This is a special type of Point structure, referred as "Structure Label". This object is different from the regular Point structure. The text label structure is used to place any desired text at any point on a geographic display. The text label behavior is like that of other types of structures, except that it cannot be selected on the display.

For more information about structure types, see [Characteristics of Structure Types](#).

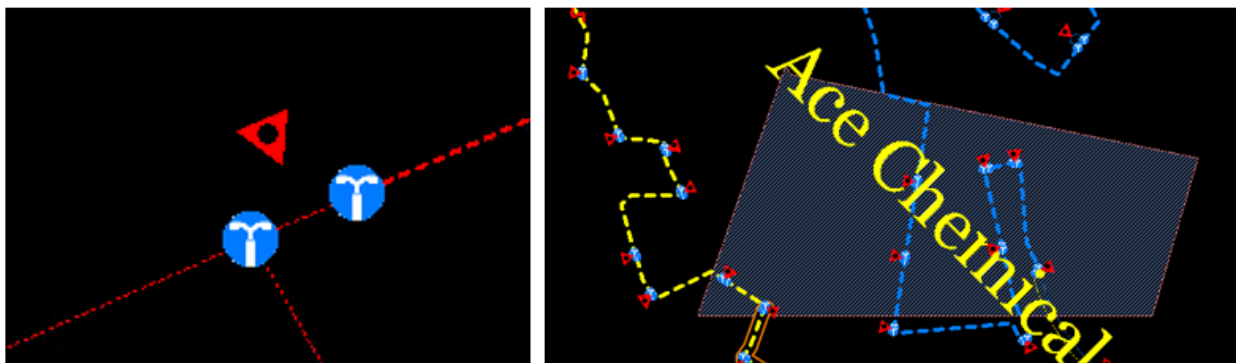


Figure 16. Point, Polygon and Text Label Structure Objects on the Network View

Modeling structures in the ADMS system is optional. You must import the structure data into the system if you want to show structures on the network view, integrate structures with DNOM objects, and show additional structure information on client displays such as analyst attribute displays.

5.1. Data Format

The structures data import process currently reads data in a *.geojson format. GeoJSON is a standard format for encoding geographic data structures (for more information, see the GeoJSON specification RFC 7946).

```
{
  "type": "FeatureCollection",
  "name": "grid_poles_0",
  "typeindex": "0",
  "crs": { "type": "name", "properties": { "name": "urn:ogc:def:crs:OGC:1.3:CRS84" } },
  "features": [
    { "type": "Feature", "properties": { "id": "1d6d0631-f23d-4b32-ada0-35alad17fb89", "name": "104419274", "owner": "FIPL", "type": "RCC", "arrangement": "A3", "spacing": "S3", "height": "104419274" }, "geometry": { "type": "Point", "coordinates": [ -87.6298, 41.8819 ] } },
    { "type": "Feature", "properties": { "id": "dbcb1c42-895e-429a-bb26-55dda33755c8", "name": "104621903", "owner": "FIPL", "type": "RCC", "arrangement": "A3", "spacing": "S1", "height": "104621903" }, "geometry": { "type": "Point", "coordinates": [ -87.6298, 41.8819 ] } }
  ]
}
```

Figure 17. GeoJSON File Format

A GeoJSON file contains the following information for each structure type (pole, streetlight, area, label, etc.):

- **typeindex:** (Required) Integer value that identifies the type of the structure. This is used to select specific data properties that are configurable in the Viewer Configuration Editor. If you do not provide a value for this parameter, the structure objects are assigned an index of 0 by default, and thus, various structure types cannot be distinguished in the ADMS client.
- **Feature:** Defines the structure and structure parameters for a specific geographic location.
- **properties:** The properties array contains key/value pairs to further describe a structure. All property values can be referenced in queries by using the property key. There are a few special properties that require specific names to be used by the system:

- **feedername:** The name of the feeder that is associated with the structure. If defined, the structure can be shown or hidden when the network view is filtered by feeder.

You can substitute the "feedername" property with "circuit_name". If substituted, any query (for example, those used in analyst attribute displays and structure search grid tabular display) must be modified to select the substitute property. Also, if you substitute this property in one structure file, you must substitute it in all structure files. If you include both properties, "feedername" is used. Only one feeder name (nominal feeder) is supported for filtering.

- **feederid:** The ID of the feeder that is associated with the structure.

You can substitute the "feederid" property with "circuit_id". If substituted, any query (for example, those used in analyst attribute displays and structure search grid tabular display) must be modified to select the substitute property. Also, if you substitute this property in one structure file, you must substitute it in all structure files. If you include both properties, "feederid" is used.

- **color:** (Applies to structure labels only) If specified, overrides the default style defined in viewer configuration.
- **font:** (Applies to structure labels only) If specified, overrides the default style configuration defined on the Structures tab of the Viewer Configuration Editor.
- **height:** (Applies to structure labels only) If specified, overrides the default style configuration defined on the Structures tab of the Viewer Configuration Editor.

- **justification:** (Applies to structure labels only) If specified, overrides the default style configuration defined on the Structures tab of the Viewer Configuration Editor.
- **rotation:** (Applies to structure labels only) If specified, overrides the default style configuration defined on the Structures tab of the Viewer Configuration Editor.
- **text:** (Applies to structure labels only) If specified, used as label text instead of the "name" property.

5.1.1. Tiles

Exporting each structure type in a single file can result in large data files that impact performance when publishing. Therefore, it is recommended to segment the structure data for a given structure type. These segments are called "tiles". Each tile covers a geographic rectangle. This also allows data updates to impact a smaller subset of files; when publishing the updates, only a small set of files are sent to ADMS clients.

5.1.2. DNOM Integration

By default, there is no direct link from DNOM objects to structure objects. Network objects, such as distribution transformers, switches, and capacitors, contain a "StructureID" field, which is populated from the descriptive data in the station XML input file. When the converter builds a station *.mod file, it adds the structure ID on which the device is mounted to the DNOM object. This value will then be available for analyst and operator attribute displays via queries.

5.2. Indexing

Before publishing *.geojson files to a client, you must index the data because:

- It allows the client to search for structures by ID or name.
- It allows the client to selectively load tiles that are required for display in each geographic window.

You can use the StructureIndexer.exe tool to index the data by creating a "Structure.index" file in the same folder as the *.geojson data files. This index file will then be sent to the clients.

To run the StructureIndexer.exe tool, you must provide a data folder containing the *.geojson files. The following conditions apply when you perform indexing:

- If the ID and Name data fields are referenced by a different name in the data, you must also provide the data fields to use for the ID and Name indexes.
- A structure ID, like a DNOM object ID, must be unique across all structures. The structure "Name" need not be unique; it can be blank.

The following table provides a list of command-line arguments supported by the Structure Indexer.

Table 1. Structure Indexer Command-Line Arguments

Argument	Description
-INPUT [PATH TO FOLDER]	Path to a directory containing one or more *.geojson files with structure objects from which index must be created.
-ID	Optional parameter. Specifies which structure object property to use as the "ID" property for index (by default, it is "id"). This value must be unique.
-NAME	Optional parameter. Specifies which structure object property to use as the "Name" property for index (by default, it is "name").
-OUTPUT	Optional parameter. Path to a directory where the Structure.index file is saved. By default, the file is saved in the Import directory.

Example:

```
D:/Eterra/Distribution/Tools/bin/StructureIndexer.exe /INPUT D:/structures /ID OBJECTID /NAME ID_NO /OUTPUT D:/structures
```

i Note

If you use non-default Name and ID properties for indexing, the Structure Search display automatically applies the overridden properties for searching and for search criteria radio button labels. However, if you want to query and return values for these properties from your GeoJSON file, you must use actual key names in queries. For example, if you use argument "/NAME ID_NO", the Structure Search display uses the "ID_NO" name for the ID radio button. However, any query such as those used for attribute popups or the Structure Search grid tabular display must query the property "ID_NO", not "NAME".

5.3. Distribution to Clients

After you index the data, you can send the *.geojson data files and the Structure.index file to ADMS clients using the File Updater tool. This tool watches for changes in the specific data files and only sends the updated files. The general process is shown in the following image.

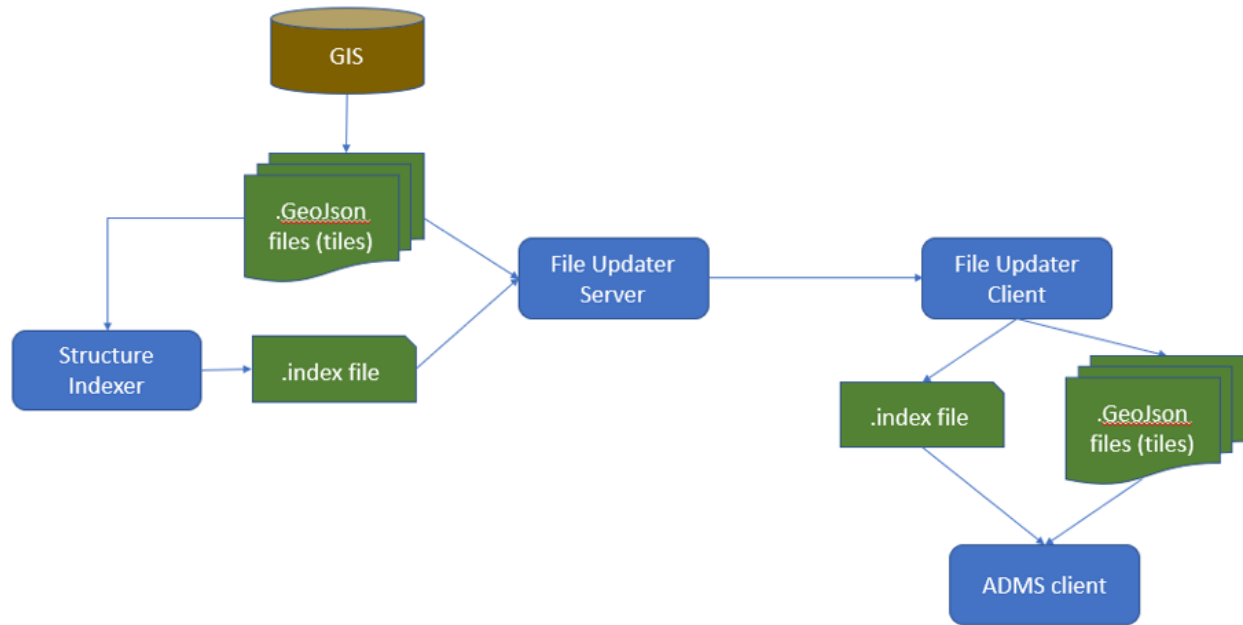


Figure 18. Structure Import Workflow

5.4. Viewer Configuration

The Viewer Configuration Editor configures how structures and other objects appear on the network view. It also configures attribute displays and toolbars. For more information, refer to the *Viewer Configuration Editor User Guide*.

5.4.1. Structure Style Parameters

On the Structures tab in the Viewer Configuration Editor, you can configure structure styles for the geographic network view display.

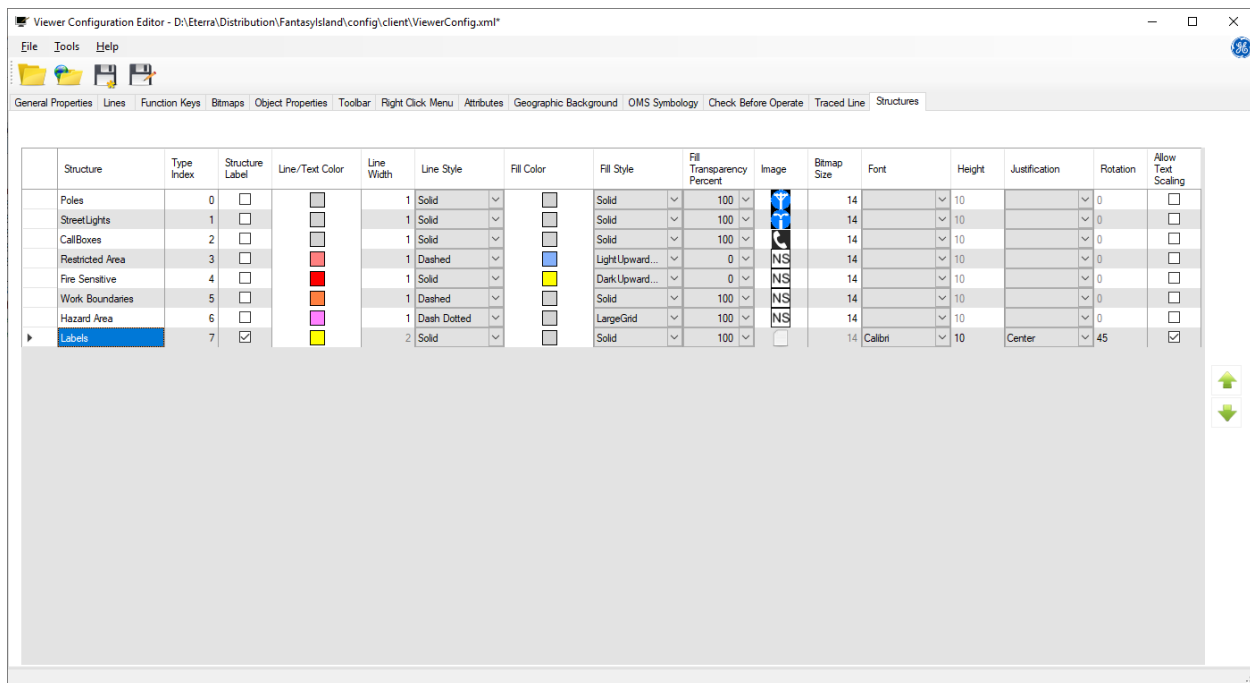


Figure 19. Configuring Structure Styles

The "Type Index" property of structures is also used on the geographic (common) toolbar to indicate which types of structures must be displayed. By default, initial display is specified in the toolbar definition, but you can choose to display specific types or all types.

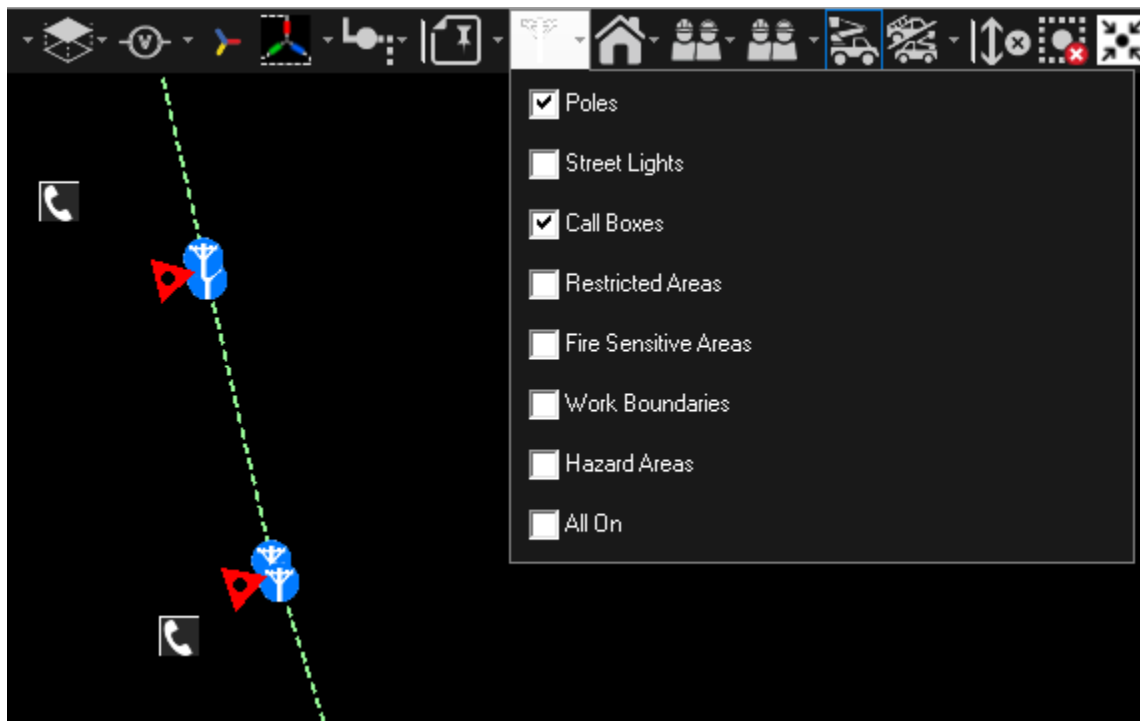


Figure 20. Geographic (Common) Toolbar

5.4.2. Attribute Displays

You can configure analyst and operator displays for the "dmsStructure" data type.

Analyst and operator attribute displays provide information that include standard fields such as %ID and %NAME as well as any property values associated with a structure. To display specific data, the key in the property data is used as a query parameter. For example, the "owner" field is retrieved as "%OWNER". The "Height" of the pole is retrieved as "%HEIGHT".

Additionally, the %TypeIndex value is also available. You can use this value to configure the information that must appear in the attribute display, such as various properties for structure types.

5.4.3. Device Toolbar

You can configure a structure toolbar using the "dmsStructure" data type.

Like the attribute displays, you can configure the structure toolbar to display the property values of a structure. The values are retrieved using the same query values as in the attribute display.

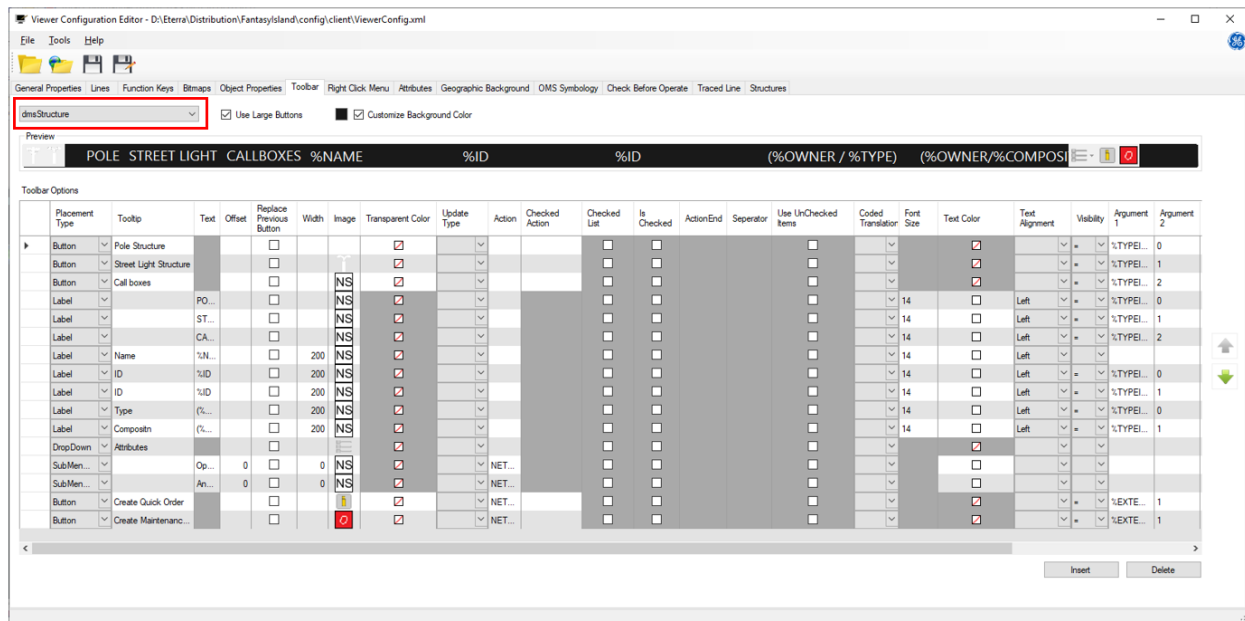


Figure 21. Toolbar Tab

You can use the %TypeIndex value in conditions to show different data for different structure types.

You can use the %ExtEnabled value in conditions to show options to create quick orders and maintenance orders for basic Point geometry structures (Poles and Streetlights in the demo model). The %ExtEnabled value is "1" (true) only for Point structures, because "extended functions" are supported only for the Point type. Extended functions include the ability to create quick orders and maintenance orders and mark structures with a common mark or mark structures as failed equipment.

5.4.4. Geographic Open Space Toolbar Menu

You can configure the structure visibility drop-down menu.

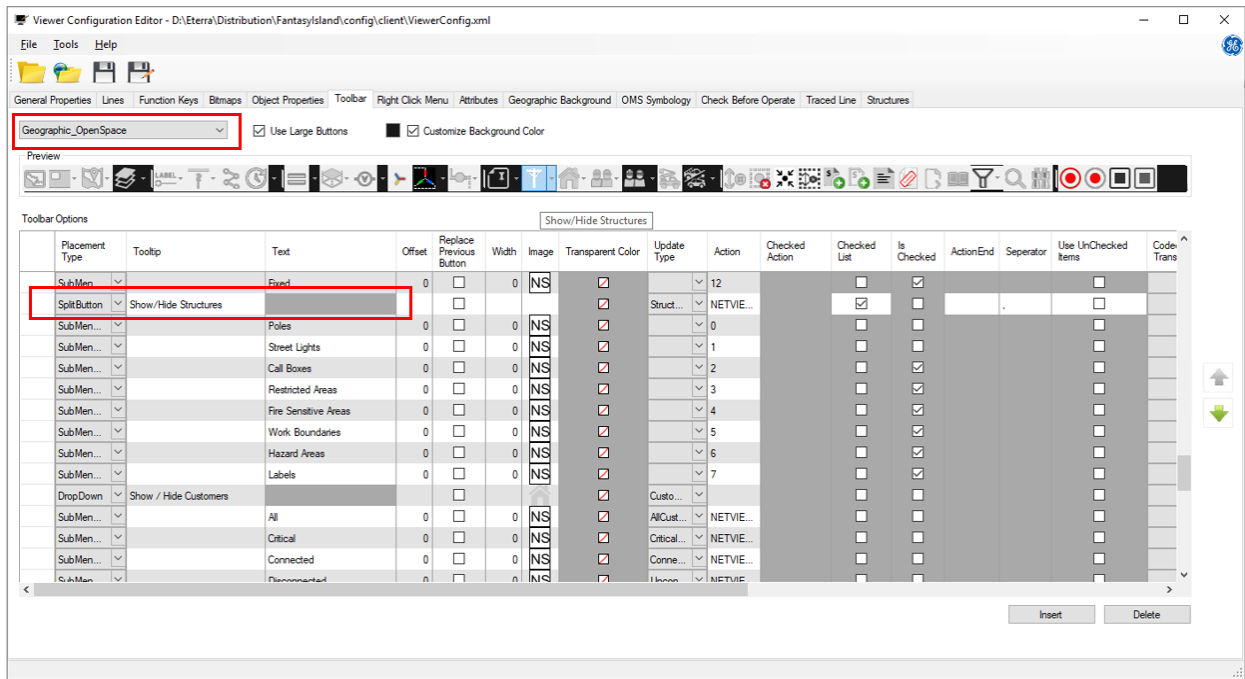


Figure 22. Geographic Open Space Toolbar Configuration

In the Show/Hide Structures drop-down menu configuration, the "Action" value of a structure type SubMenu is equal to StructureIndex. The "Is Checked" option specifies whether a structure type is shown by default when the network view is opened. (To modify "Is Checked" for structure types, you must temporarily uncheck the "Checked List" box for the parent Show/Hide Structures SplitButton.)

5.5. Structure Search Display

Like the attribute displays, you can configure the Structure Search grid tabular display to show property values of a structure. The values are retrieved using the same query values as in the attribute display.

Note

Slash is not supported when searching for a Structure.

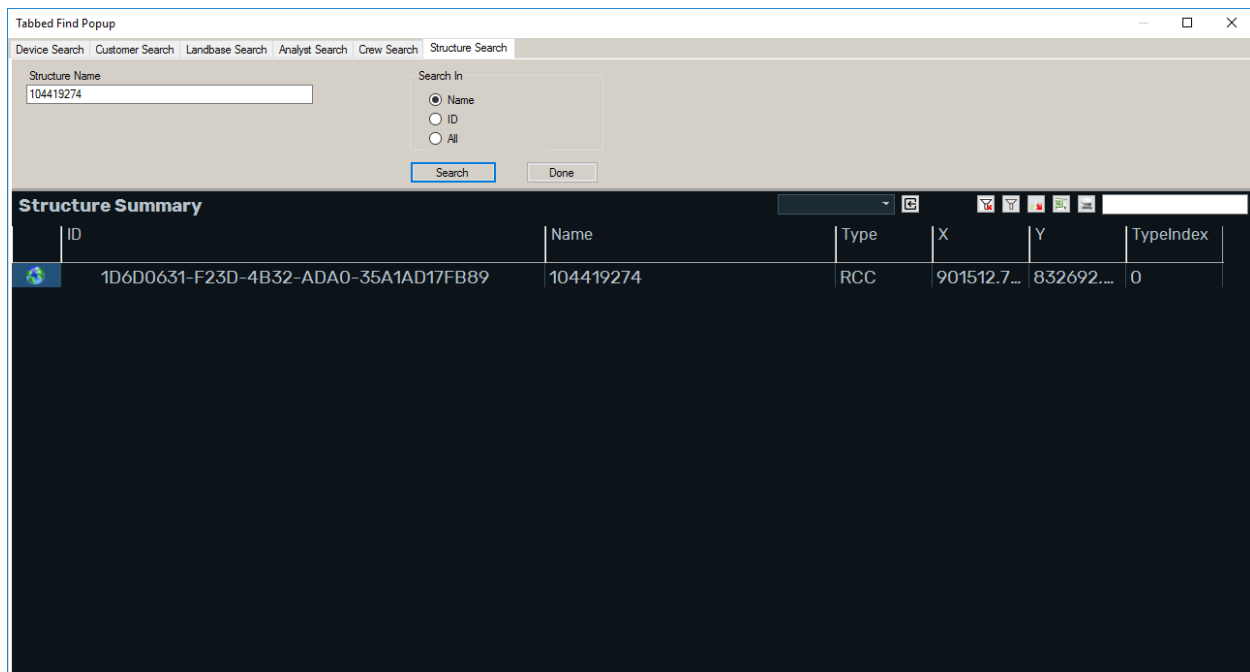


Figure 23. Structure Search Display

5.6. Characteristics of Structure Types

Structure object behavior depends on the type of GeoJSON geometry the structure utilizes.

- Point (single point)
 - Drawn using a bitmap placed at the structure's single geographic coordinates.
 - The "location" of the structure is simply the point coordinates.
 - Selectable. You can select the structure by clicking the single point location of the structure.
 - Supports filtering by feeder. One static feeder name, as defined by the "feedername" property in the GeoJSON file, is supported for filtering.
 - Supports extended functions. You can create quick orders and maintenance orders. You can mark structures with a common mark or mark structures as failed equipment.
- MultiPoint (collection of points)
 - Drawn using a bitmap placed at each of the structure's geographic point coordinates (same bitmap at each coordinate).
 - The "location" of the structure is the point closest to the center between all points.
 - Selectable. You can select the structure by clicking any single point location of the structure.
 - Supports filtering by feeder. One static feeder name, as defined by the "feedername" property in the GeoJSON file, is supported for filtering.
 - Does not support extended functions.

LineString (segmented line)

- Drawn as a segmented line using styles defined in the viewer configuration.
- The "location" of the structure is the center of the line.
- Selectable. You can select the structure by clicking anywhere on the line.
- Supports filtering by feeder. One static feeder name, as defined by the "feedername" property in the GeoJSON file, is supported for filtering.
- Does not support extended functions.
- MultiLineString (collection of segmented lines)
 - Drawn as multiple segmented lines using styles defined in the viewer configuration.
 - The "location" of the structure is the center of the line closest to the center between all lines.
 - Selectable. You can select the structure by clicking anywhere on any of the lines.
 - Supports filtering by feeder. One static feeder name, as defined by the "feedername" property in the GeoJSON file, is supported for filtering.
 - Does not support extended functions.
- Polygon (single polygon)
 - Drawn as a polygon using styles defined in the viewer configuration.
 - The "location" of the structure is the centroid of the polygon.
 - Selectable
 - If filled, you can select the structure by clicking anywhere within the polygon.
 - If unfilled, you can select the structure by clicking anywhere on the polygon perimeter.
 - Supports filtering by feeder. One static feeder name, as defined by the "feedername" property in the GeoJSON file, is supported for filtering.
 - Does not support extended functions.
- MultiPolygon (collection of polygons)
 - Drawn as multiple polygons using styles defined in the viewer configuration.
 - The "location" of the structure is the centroid of the polygon closest to the center between all polygons.
 - Selectable
 - If filled, you can select the structure by clicking anywhere within any polygon.
 - If unfilled, you can select the structure by clicking anywhere on any polygon perimeter.
 - Supports filtering by feeder. One static feeder name, as defined by the "feedername" property in the GeoJSON file, is supported for filtering.
 - Does not support extended functions.
- Point (Text Label)

- Specialized type of a single point (Point) structure
- Drawn as text using styles defined in the viewer configuration. The following style settings can be overridden by GeoJSON properties for each feature:
 - color (RGB integer)
 - font (name of font family)
 - height (pixels at 1:1 zoom)
 - justification ("Left", "Center", "Right")
 - rotation (0359 degrees)

Note

Overriding styles for each structure within the GeoJSON file may have a performance impact. Be cautious when overriding default styles for structures.

- The "location" of the structure is simply the point coordinates.
- Not selectable. It is assumed that the required function is only to show the label text.
- Supports filtering by feeder. One static feeder name, as defined by the "feedername" property in the GeoJSON file, is supported for filtering.
- Does not support extended functions.

5.7. Modeling Equipment and Structures

5.7.1. Data Format

The equipment and structure index creation currently reads data in a *.geojson format. GeoJSON is a standard format for encoding geographic data structures (for more information, see the GeoJSON specification RFC 7946). In addition to the structure information, the *.geojson files also needs to include the DNOM objects associated with the structure.

```
{
  "type": "FeatureCollection",
  "name": "grid_poles_1",
  "typeindex": "0",
  "crs": { "type": "name", "properties": { "name": "urn:ogc:def:crs:OGC:1.3:CRS84" }},
  "features": [
    {
      "type": "Feature", "geometry": { "type": "Point", "coordinates": [ 897326.6875, 822417.125 ] }, "properties": { "id": "897ae4f1-47ed-44b9-aad7-4939f2185880", "name": "115759868", "owner": "FI",
      "dnomids": [ { "dnomid": "338773102", "dnomname": "338773084", "dnomphase": "CN", "feederid": "8154782", "feedername": "F8664", "substation": "COBO", "deviceType": "Line", { "dnomid": "115759874", "dnomname": "1"
    }
  ]
}
```

Figure 24. GeoJSON File Format

Structures are defined in the "features" array. The structure information is defined in the "properties" collection. The specific structure is identified with the following two properties:

- **id:** The ID of the structure.
- **name:** The name of the structure.

A structure can be associated for none, one, or more than one piece of equipment. This association

is described in the "dnomids" array for each feature.

If the "dnomids" entry is absent or empty, the compiler will assume that the structure is not associated with any equipment.

Equipment associated with a structure is defined by the following list of entries:

- **dnomid**: The DNOM ID for the equipment.
- **dnomname**: The name of the equipment.
- **dnomphase**: The phase of the equipment.
- **feederid**: The ID of the feeder for which this equipment is a part.
- **feedername**: The name of the feeder for which this equipment is a part.
- **substation**: The station name for which this equipment is part.
- **deviceType**: The device type of the equipment. This value can be one of the following:
 - AutoTransformer
 - Breaker
 - Bus
 - Capacitor
 - CompositeSwitch
 - DistributionTransformer
 - Elbow
 - EnergyResource
 - FaultDetector
 - Fuse
 - Generator
 - Jumper
 - Line
 - Load
 - NetworkProtector
 - PowerTransformer
 - PrimaryMeter
 - Recloser
 - Regulator
 - Sectionalizer
 - SeriesReactor
 - StepTransformer
 -

- Switch
 - SwitchCabinet
 - TransformerBank
 - UIContainer
 - Vault
 - ZigZagTransformer

Example:

```
"features" : [
{
  "type": "Feature",
  "geometry": {"type": "Point", "coordinates": [697326.6875, 722417.225 ]},
  "properties": {
    "id" : "123ae4f1-4567-44b9-1234-4839f2185880",
    "name" : "Pole-123",
    ...
  },
  "dnomids" : [
  {
    "dnomid": "12345",
    "dnomname": "LineABC",
    "dnomphase": "ABCN",
    "feederid": "567",
    "feedername": "F567",
    "substation": "SAMPLE",
    "deviceType": "Line"
  }
  ]
},
...
]
```

5.7.2. Indexing

Before building the model files with the converter, create the equipment and structure index file for the following reasons:

- It allows the converter to associate the structures with equipment and set up the halos for equipment sharing structures with equipment from different feeders.
- The index is required by the StructureServer to provide the information for the Structure Information popup displays.

You can use the StructureCompiler.exe tool to create the EquipmentAndStructures.index file, which is placed on the server and used by the StructureServer.

To run the StructureCompiler.exe tool, you must provide a data folder containing the *.geojson files, as well as the output file. The following conditions apply when you create the index file:

- The *.geojson files must contain the DNOM objects associated with each structure. If no

association is found on a structure, it assumes that the structure is not associated with any equipment.

- If required, the index file can be encrypted.

Table 2. Structure Compiler Command-Line Arguments

Argument	Description
/INPUT	Path to the directory containing one or more "*.geojson" files with structure object and the associated equipment objects to create the equipment and structure index. The index is used by the Converter tool as well as the StructureServer.
/OUTPUT	The file name of the equipment and structure index file that will be created.
/ENCMODE	Example: StructureCompiler /INPUT D:\Structures /OUTPUT D:\Structures\EquipmentAndStructures.index
/ENCKEYSCOPE	Defines if the resulting index must be encrypted or not. The value can be "Encrypted" or "None". By default this is assumed to be "None". This parameter is only required if the index must be encrypted.
/ENCKEYVERSION	The encryption scope. The format for this value is an uppercase text string with maximum length of 11 characters. This parameter is required if the index must be encrypted.
/ENCKEYSTORAGE	The encryption key storage. This value can either be: • Default: The value internally resolved to EnviromentVariableDpApi • EnvironmentVariable: Specifies that the encryption key is defined in the system using the environment variable and the key value is not encrypted • EnvironmentVariableDpApi: Specifies that the encryption key is defined in the system using an environment variable and the key is encrypted using the DPAPI encryption method This parameter is required if the index must be encrypted.

Example:

```
D:\Eterra\Distribution\Tools\bin\StructureCompiler.exe /INPUT D:\Structures /OUTPUT
D:\Structures\EquipmentAndStructures.index
```

6. Building the Postal_Inventory.xml File

The Postal_Inventory.xml file is used to map postal (ZIP) codes to towns.

The minimum content required for the file is:

```
<?xml version="1.0"?>
<POSTAL_ZIPS>
</POSTAL_ZIPS>
```

If the Postal_inventory.shp and AddrRg.shp files exist as part of the TeleAtlas GIS data, they can be used to build a complete Postal_Inventory.xml file with the Geographic Background Compiler. The Postal_Inventory.shp and AddrRg.shp files should be placed in the same top-level folder as the rest of the GIS data folders.

1. On the Paths tab of the Configuration window of the Geographic Background Compiler:
 - For the Search File Name entry, browse to create the full path to the AddrRg.shp file.
 - For the Postal Inventory File entry, browse to create the full path to the Postal_Inventory.shp file.
2. Select the Compile Search (when Compile button is pressed) check box.
3. Ensure that the Input Folder path is pointed to the top-level folder for all the GIS subfolders.
4. On the Compiler tab, click the Compile button.

When compilation is complete, the Postal_Inventory.xml file will be in the specified output folder, along with the compiled .geo files for each tile.

The Postal_Inventory.xml file should be placed in the same folder where all the background files are located. The background path is defined in the Net Dynamics Configuration Editor, as described in [Defining Background Parameters in the Net Dynamics Configuration Editor](#).

7. The AddressLookup.xml File

The AddressLookup.xml file is used as a lookup table to normalize addresses.

The minimum content required for the file is:

```
<?xml version="1.0"?>
<Tables>
</Tables>
```

The AddressLookup.xml file is organized into four major tables: State Lookup, Direction Lookup, Street Type Lookup, and Secondary Unit Lookup. The structure is:

```
<Tables>
  <!-- state -->
  <Table src="state_lookup">
  </Table>
  <!-- direction_lookup -->
  <Table src="direction_lookup">
  </Table>
  <!-- street_type_lookup -->
  <Table src="street_type_lookup">
  </Table>
  <!-- secondary_unit_lookup -->
  <Table src="secondary_unit_lookup">
  </Table>
</Tables>
```

The data within each major table is organized as rows, each row populated with two words: the source word and the resolved word. For example, in the Direction Lookup table, entries for "southwest" might include the following:

```
<Row>
  <Word>SW</Word>
  <Word>SW</Word>
</Row>
<Row>
  <Word>SOUTH-WEST</Word>
  <Word>SW</Word>
</Row>
<Row>
  <Word>SOUTHWEST</Word>
  <Word>SW</Word>
</Row>
```

If the user types "southwest" or "south-west", either is resolved to "SW".

The Street Type table can include potential abbreviations and misspellings for Avenue, Boulevard, Place, Circle, etc.

The Secondary Unit table can include common indicators for Apartment, Building, Floor, Bay,

Department, etc.

The AddressLookup.xml file should be placed in the same folder where all the background files are located. The background path is defined in the Net Dynamics Configuration Editor, as described in [Defining Background Parameters in the Net Dynamics Configuration Editor](#).

8. Displaying the Landbase Search Tab

For the Landbase Search tab to appear in the Geographic Background Compiler and on the Tabbed Find Popup in the client, the BackgroundSearchFile.gsch file must be created (as detailed in [Options Tab](#)) and placed in an accessible folder on the client computer.

1. On the client computer, click Start > Programs > Eterra > Distribution > Tools > Configuration Tools.

The Net Dynamics Configuration Editor opens.

2. Open the correct configuration file for the client installation and click the Backgrounds Editor button.
3. Clear the Background File field.
4. In the Background Path field, enter the path to the BackgroundSearchFile.gsch file.
5. Set both "Background Load?" and "Use ShapeTiles?" to True.
6. In the Search File field, enter BackgroundSearchFile.gsch.
7. Save the file and start the client.

When the Tabbed Find Popup is displayed, the Landbase Search tab is available.

The AddressLookup.xml and Postal_Inventory.xml files must be placed in the same folder where all the background files are located. The background path is defined in the Net Dynamics Configuration Editor, as described in [Defining Background Parameters in the Net Dynamics Configuration Editor](#).

9. Landbase Layer Types

There are four main categories or "layers" of land-based background information: water, footprints, roads, and boundaries. Each of these main categories is further broken down into four sublevels numbered 1 through 4. The purpose of this is to allow each of the four sublevels to be visible at different zoom levels.

For example, the water levels can be broken down as follows:

- **water1:** Major water such as oceans.
- **water2:** Large lakes and rivers.
- **water3:** Intermediate lakes and rivers.
- **water4:** Minor waterways (such as a neighbor's pool or a creek).

Although the categories are not configurable, they can be used for information other than what they strictly state. For example, the Boundaries4 layer can be used to show railroad tracks.

The Viewer Configuration Editor (VCE) is used to define the zoom-based visibility of these various layers. The table below shows the existing correlations between the two lists.

Table 3. Correlation Between Landbase and VCE Layers

Landbase Layers	VCE Layers
Water1	Major water
Water2	Large water
Water3	Intermediate water
Water4	Minor water
Building Footprints1	Plants, international airports, and ports
Building Footprints2	Substations, universities, and minor airports
Building Footprints3	Landmarks and large parks
Building Footprints4	Footprints, customers, and other
Roads1	Major roads
Roads2	Large roads
Roads3	Intermediate roads
Roads4	Minor roads
Boundaries1	Areas and counties
Boundaries2	SVC and cities
Boundaries3	Map boundaries
Boundaries4	Railroads